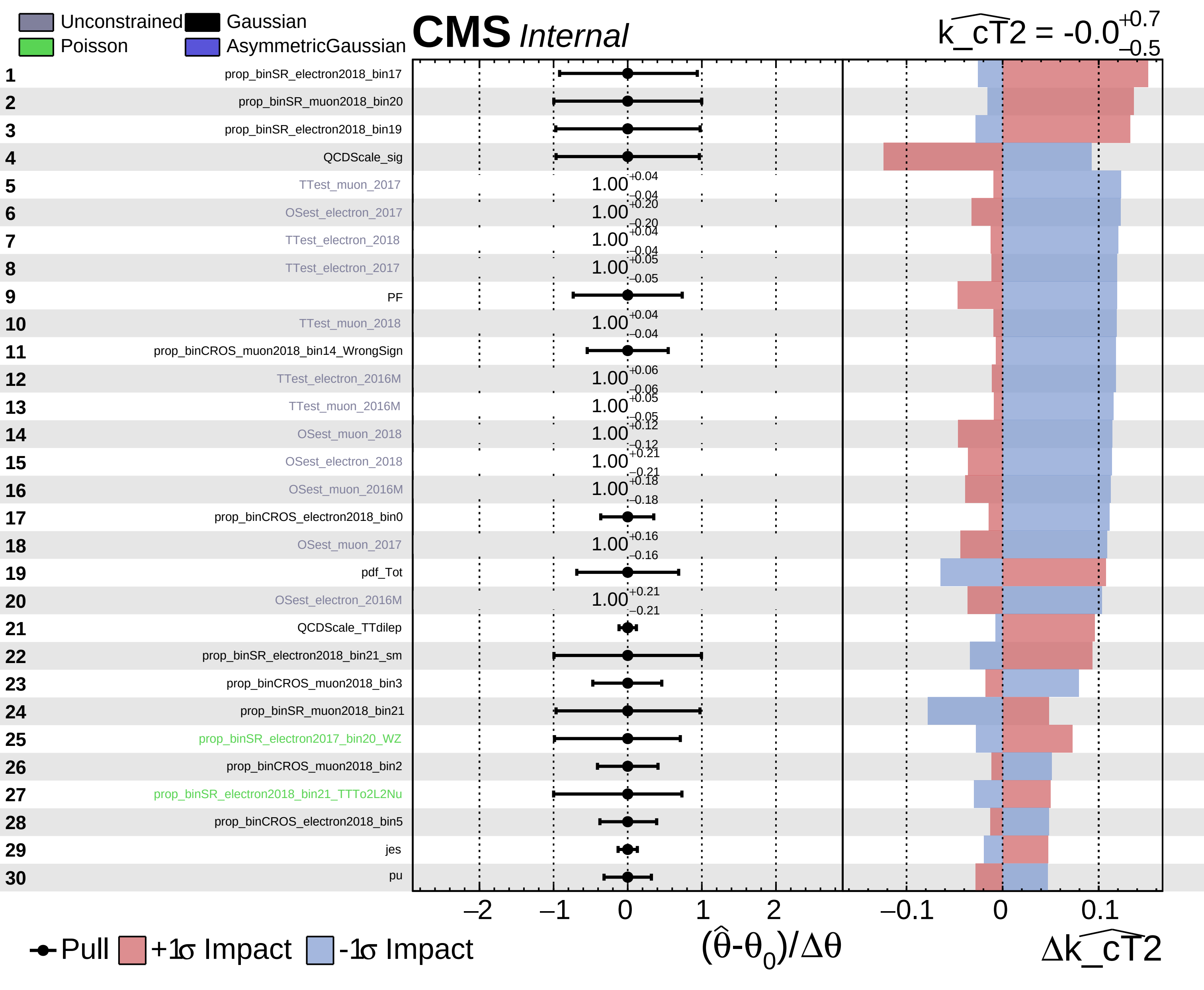
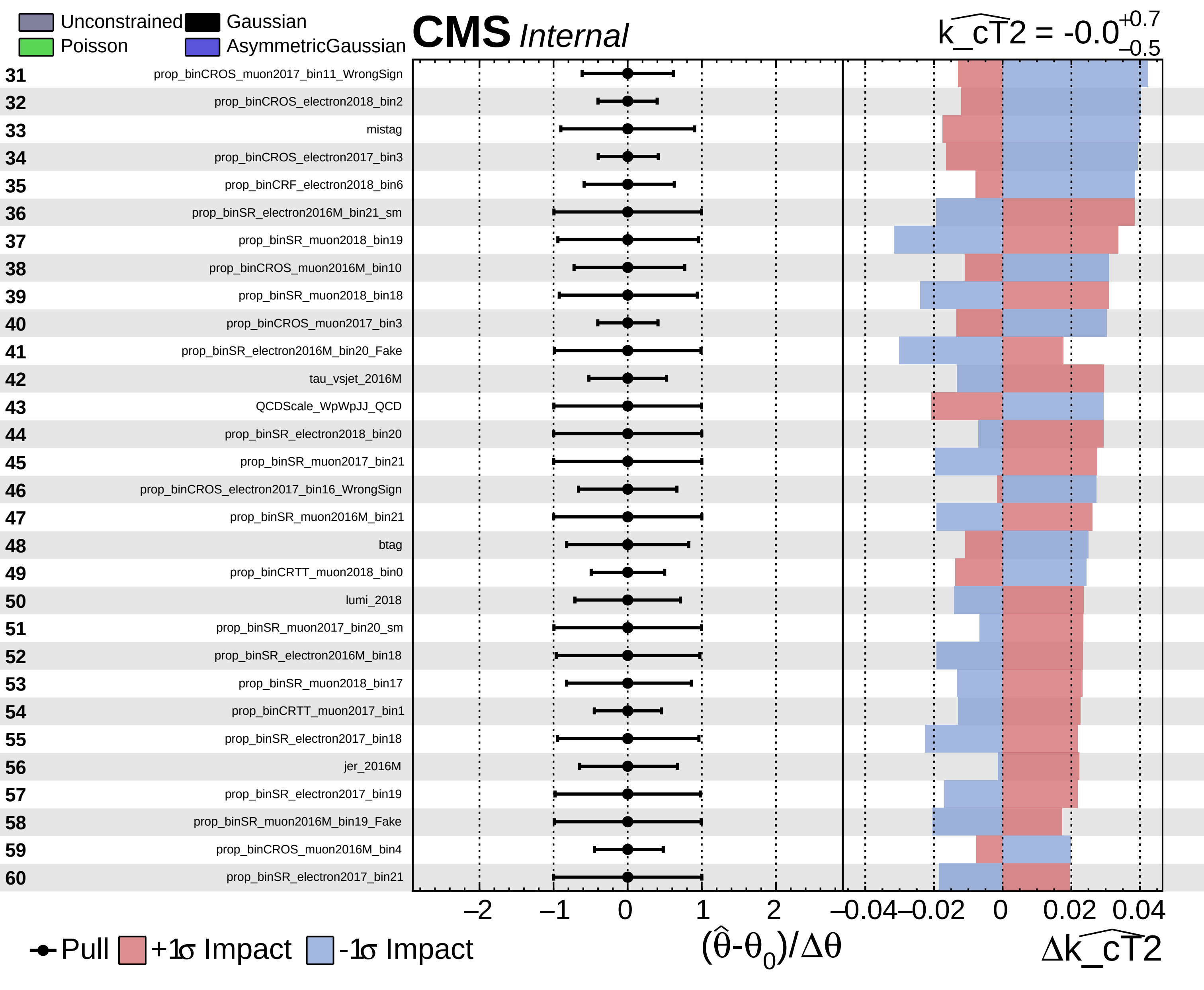
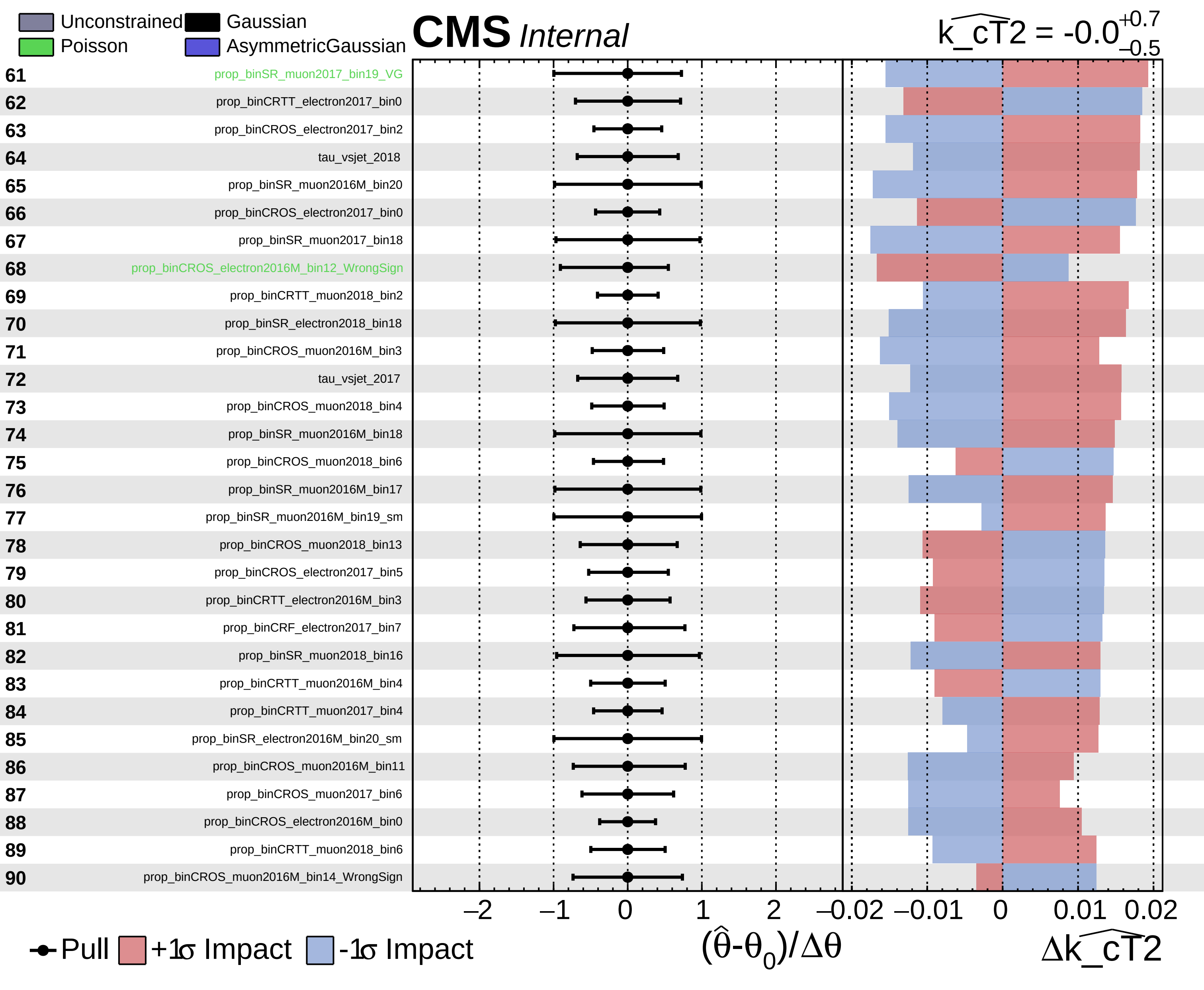
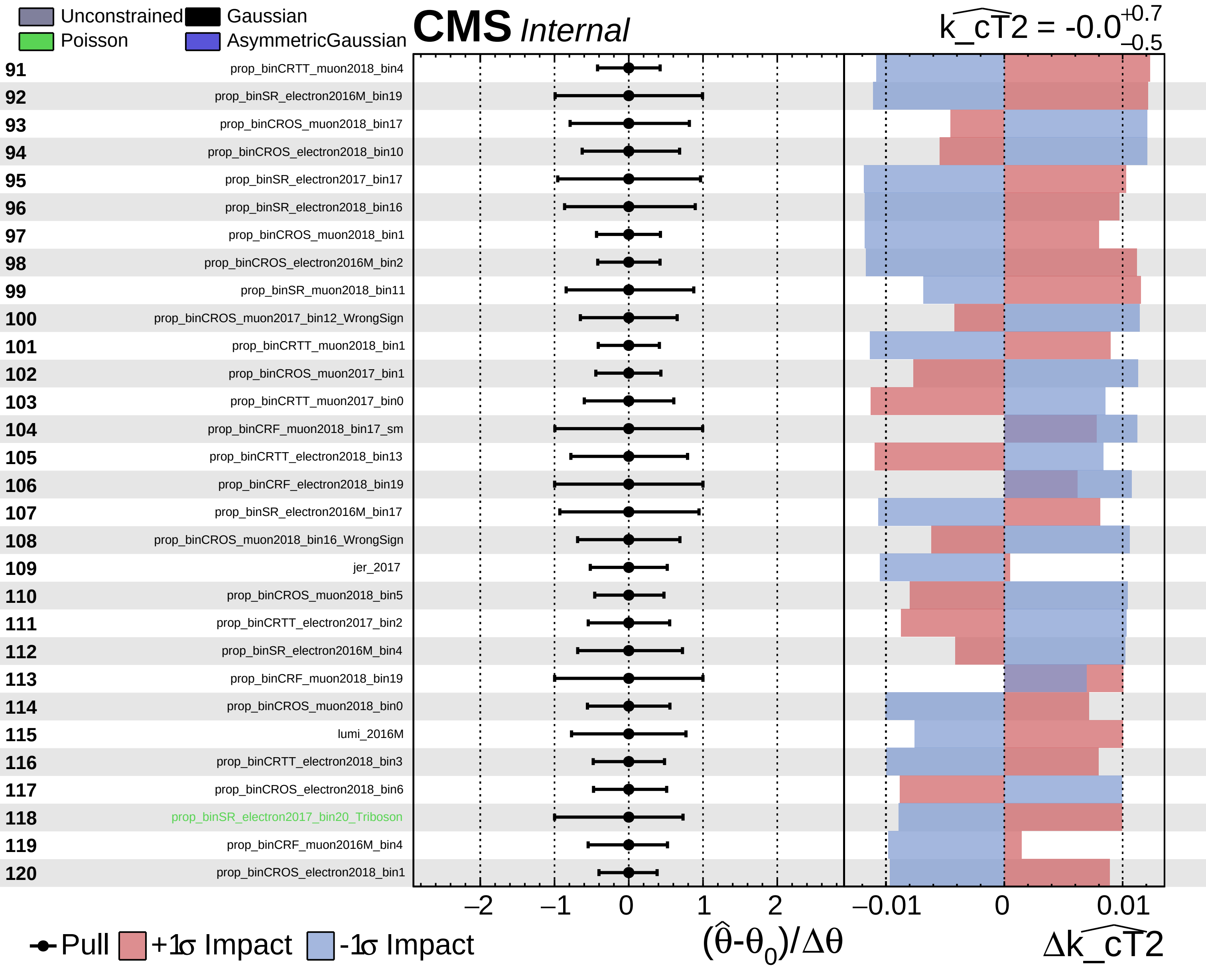


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+0.7}_{-0.5}$
1 prop_binSR_electron2018_bin17
2 prop_binSR_muon2018_bin20
3 prop_binSR_electron2018_bin19
4 QCDScale_sig
5 TTest_muon_2017 $1.00^{+0.04}_{-0.04}$
6 OSest_electron_2017 $1.00^{+0.20}_{-0.20}$
7 TTest_electron_2018 $1.00^{+0.04}_{-0.04}$
8 TTest_electron_2017 $1.00^{+0.05}_{-0.05}$
9 PF
10 TTest_muon_2018 $1.00^{+0.04}_{-0.04}$
11 prop_binCROS_muon2018_bin14_WrongSign
12 TTest_electron_2016M $1.00^{+0.06}_{-0.06}$
13 TTest_muon_2016M $1.00^{+0.05}_{-0.05}$
14 OSest_muon_2018 $1.00^{+0.12}_{-0.12}$
15 OSest_electron_2018 $1.00^{+0.21}_{-0.21}$
16 OSest_muon_2016M $1.00^{+0.18}_{-0.18}$
17 prop_binCROS_electron2018_bin0
18 OSest_muon_2017 $1.00^{+0.16}_{-0.16}$
19 pdf_Tot
20 OSest_electron_2016M $1.00^{+0.21}_{-0.21}$
21 QCDScale_TTdilep
22 prop_binSR_electron2018_bin21_sm
23 prop_binCROS_muon2018_bin3
24 prop_binSR_muon2018_bin21
25 prop_binSR_electron2017_bin20_WZ
26 prop_binCROS_muon2018_bin2
27 prop_binSR_electron2018_bin21_TTTo2L2Nu
28 prop_binCROS_electron2018_bin5
29 jes
30 pu
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

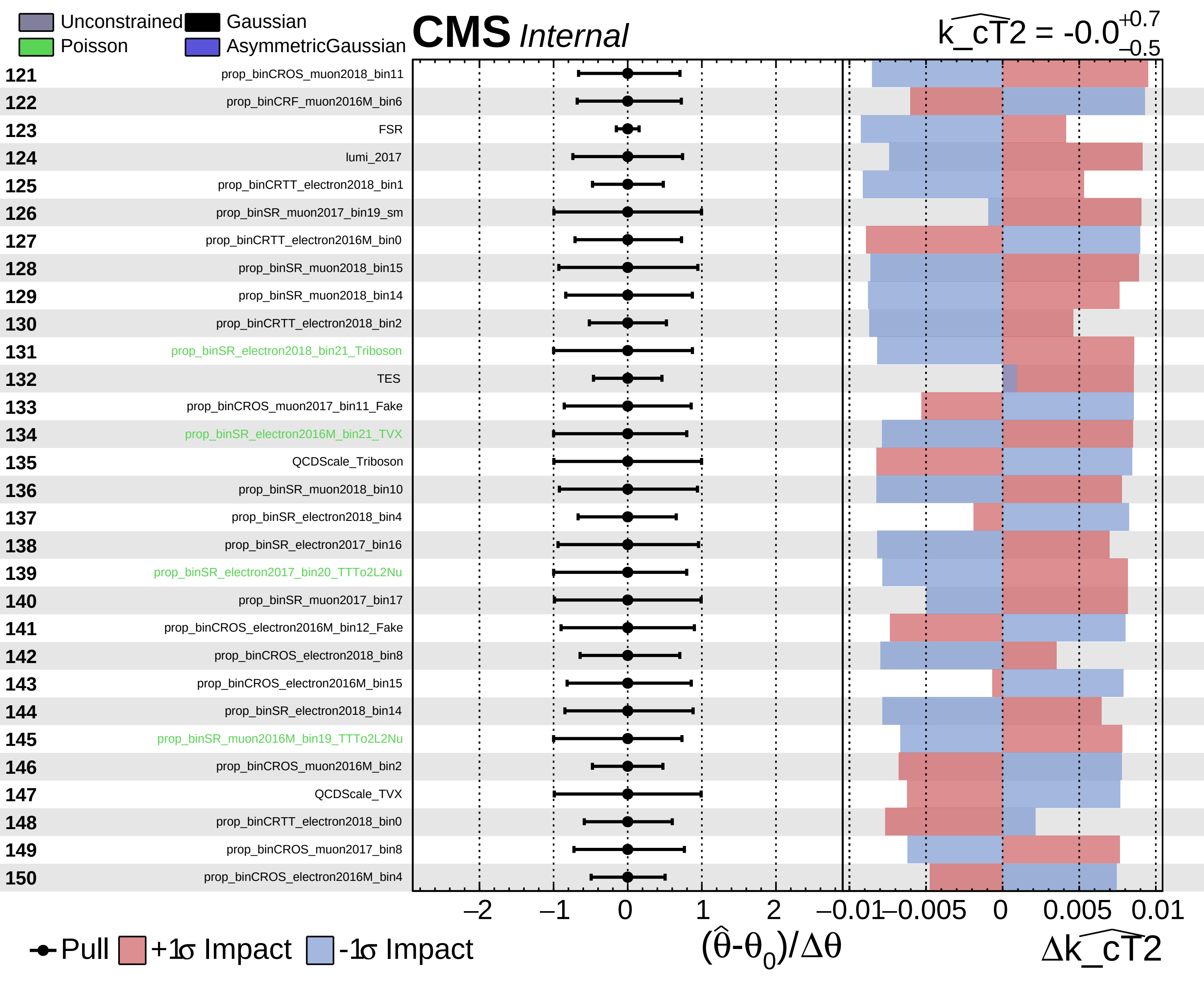


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
31 prop_binCROS_muon2017_bin11_WrongSign
32 prop_binCROS_electron2018_bin2
33 mistag
34 prop_binCROS_electron2017_bin3
35 prop_binCRF_electron2018_bin6
36 prop_binSR_electron2016M_bin21_sm
37 prop_binSR_muon2018_bin19
38 prop_binCROS_muon2016M_bin10
39 prop_binSR_muon2018_bin18
40 prop_binCROS_muon2017_bin3
41 prop_binSR_electron2016M_bin20_Fake
42 tau_vsjet_2016M
43 QCDScale_WpWpJJ_QCD
44 prop_binSR_electron2018_bin20
45 prop_binSR_muon2017_bin21
46 prop_binCROS_electron2017_bin16_WrongSign
47 prop_binSR_muon2016M_bin21
48 btag
49 prop_binCRTT_muon2018_bin0
50 lumi_2018
51 prop_binSR_muon2017_bin20_sm
52 prop_binSR_electron2016M_bin18
53 prop_binSR_muon2018_bin17
54 prop_binCRTT_muon2017_bin1
55 prop_binSR_electron2017_bin18
56 jer_2016M
57 prop_binSR_electron2017_bin19
58 prop_binSR_muon2016M_bin19_Fake
59 prop_binCROS_muon2016M_bin4
60 prop_binSR_electron2017_bin21
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

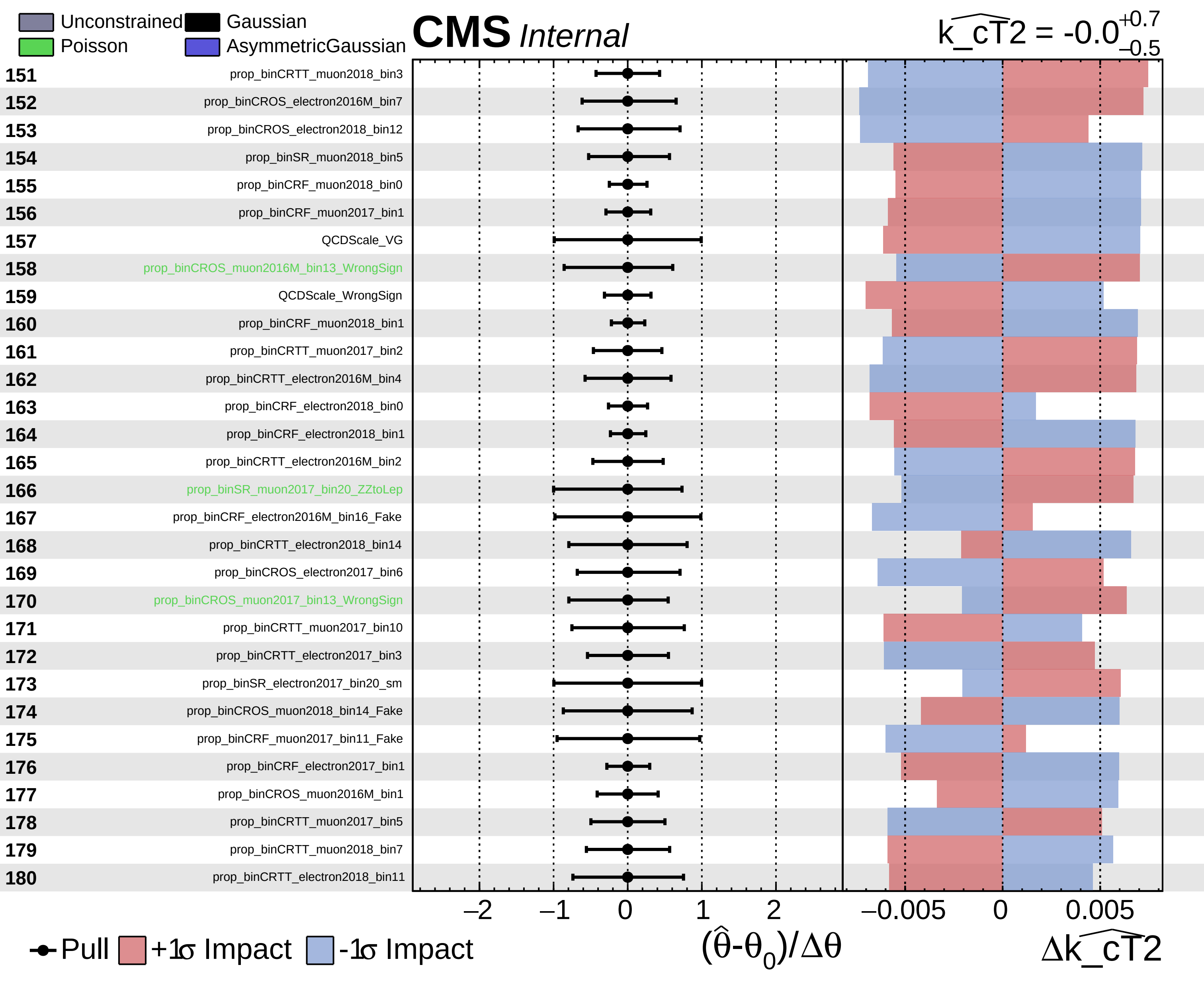

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
61 prop_binSR_muon2017_bin19_VG
62 prop_binCRTT_electron2017_bin0
63 prop_binCROS_electron2017_bin2
64 tau_vsjet_2018
65 prop_binSR_muon2016M_bin20
66 prop_binCROS_electron2017_bin0
67 prop_binSR_muon2017_bin18
68 prop_binCROS_electron2016M_bin12_WrongSign
69 prop_binCRTT_muon2018_bin2
70 prop_binSR_electron2018_bin18
71 prop_binCROS_muon2016M_bin3
72 tau_vsjet_2017
73 prop_binCROS_muon2018_bin4
74 prop_binSR_muon2016M_bin18
75 prop_binCROS_muon2018_bin6
76 prop_binSR_muon2016M_bin17
77 prop_binSR_muon2016M_bin19_sm
78 prop_binCROS_muon2018_bin13
79 prop_binCROS_electron2017_bin5
80 prop_binCRTT_electron2016M_bin3
81 prop_binCRF_electron2017_bin7
82 prop_binSR_muon2018_bin16
83 prop_binCRTT_muon2016M_bin4
84 prop_binCRTT_muon2017_bin4
85 prop_binSR_electron2016M_bin20_sm
86 prop_binCROS_muon2016M_bin11
87 prop_binCROS_muon2017_bin6
88 prop_binCROS_electron2016M_bin0
89 prop_binCRTT_muon2018_bin6
90 prop_binCROS_muon2016M_bin14_WrongSign
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

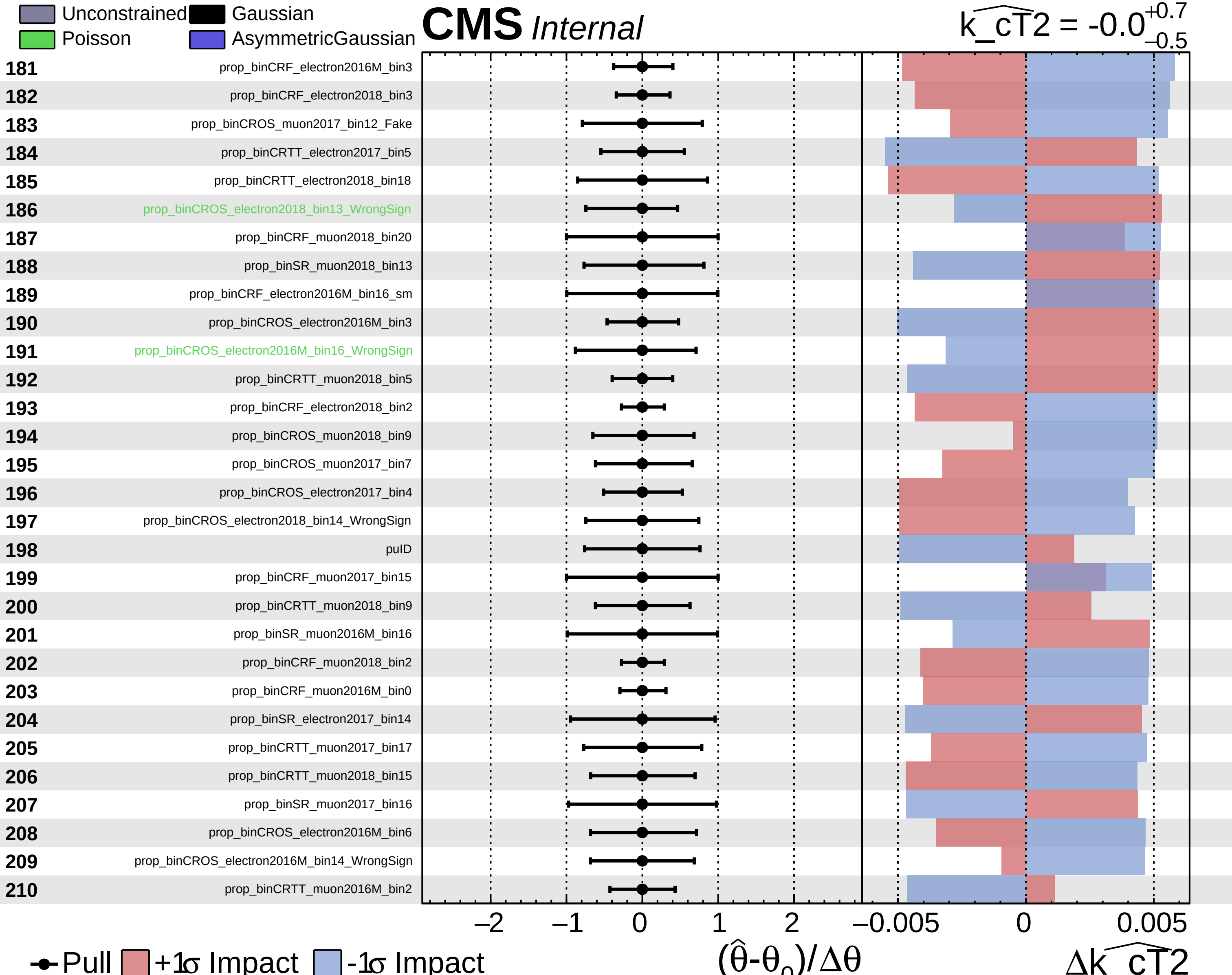


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
91 prop_binCRTT_muon2018_bin4
92 prop_binSR_electron2016M_bin19
93 prop_binCROS_muon2018_bin17
94 prop_binCROS_electron2018_bin10
95 prop_binSR_electron2017_bin17
96 prop_binSR_electron2018_bin16
97 prop_binCROS_muon2018_bin1
98 prop_binCROS_electron2016M_bin2
99 prop_binSR_muon2018_bin11
100 prop_binCROS_muon2017_bin12_WrongSign
101 prop_binCRTT_muon2018_bin1
102 prop_binCROS_muon2017_bin1
103 prop_binCRTT_muon2017_bin0
104 prop_binCRF_muon2018_bin17_sm
105 prop_binCRTT_electron2018_bin13
106 prop_binCRF_electron2018_bin19
107 prop_binSR_electron2016M_bin17
108 prop_binCROS_muon2018_bin16_WrongSign
109 jer_2017
110 prop_binCROS_muon2018_bin5
111 prop_binCRTT_electron2017_bin2
112 prop_binSR_electron2016M_bin4
113 prop_binCRF_muon2018_bin19
114 prop_binCROS_muon2018_bin0
115 lumi_2016M
116 prop_binCRTT_electron2018_bin3
117 prop_binCROS_electron2018_bin6
118 prop_binSR_electron2017_bin20_Triboson
119 prop_binCRF_muon2016M_bin4
120 prop_binCROS_electron2018_bin1
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT2

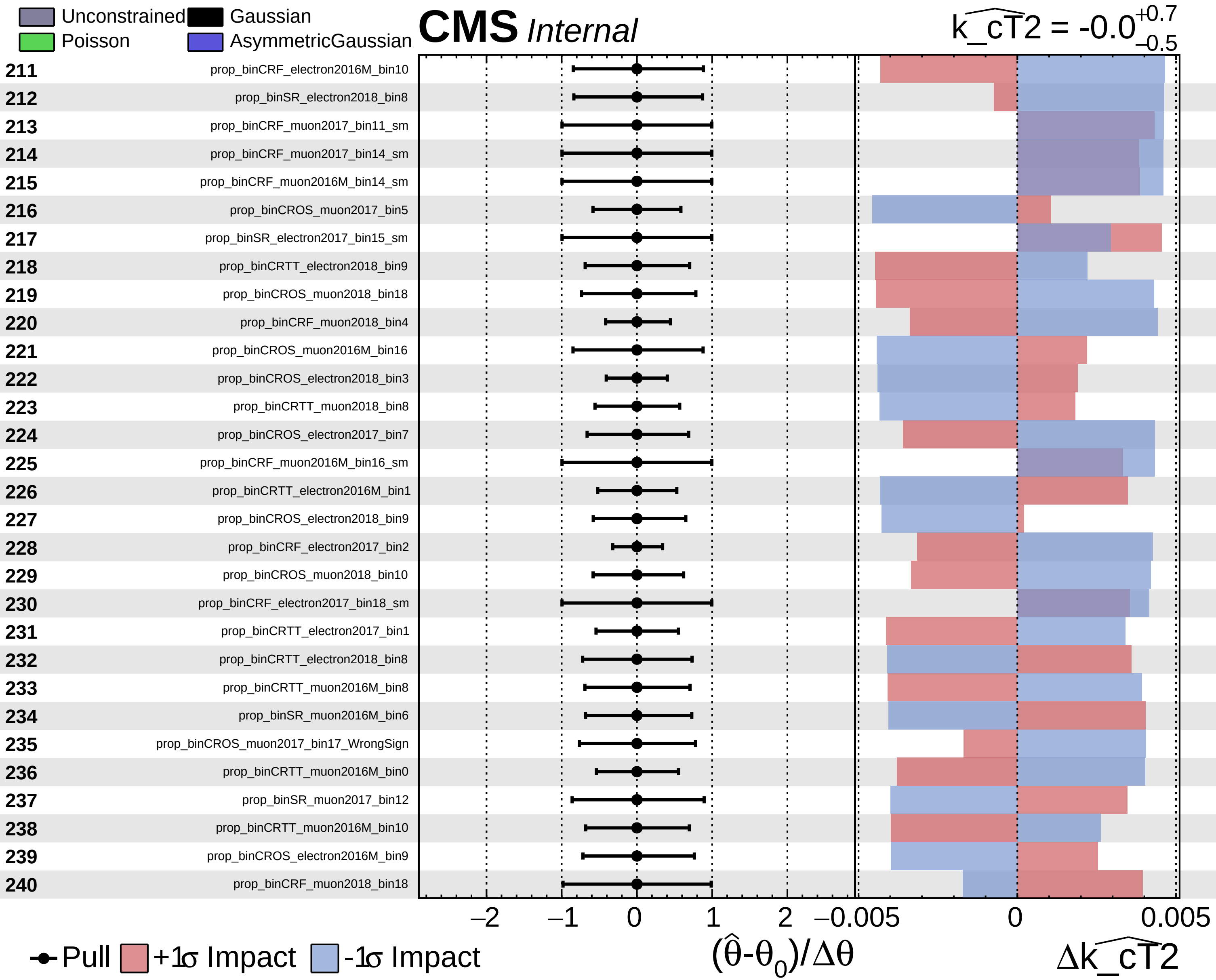


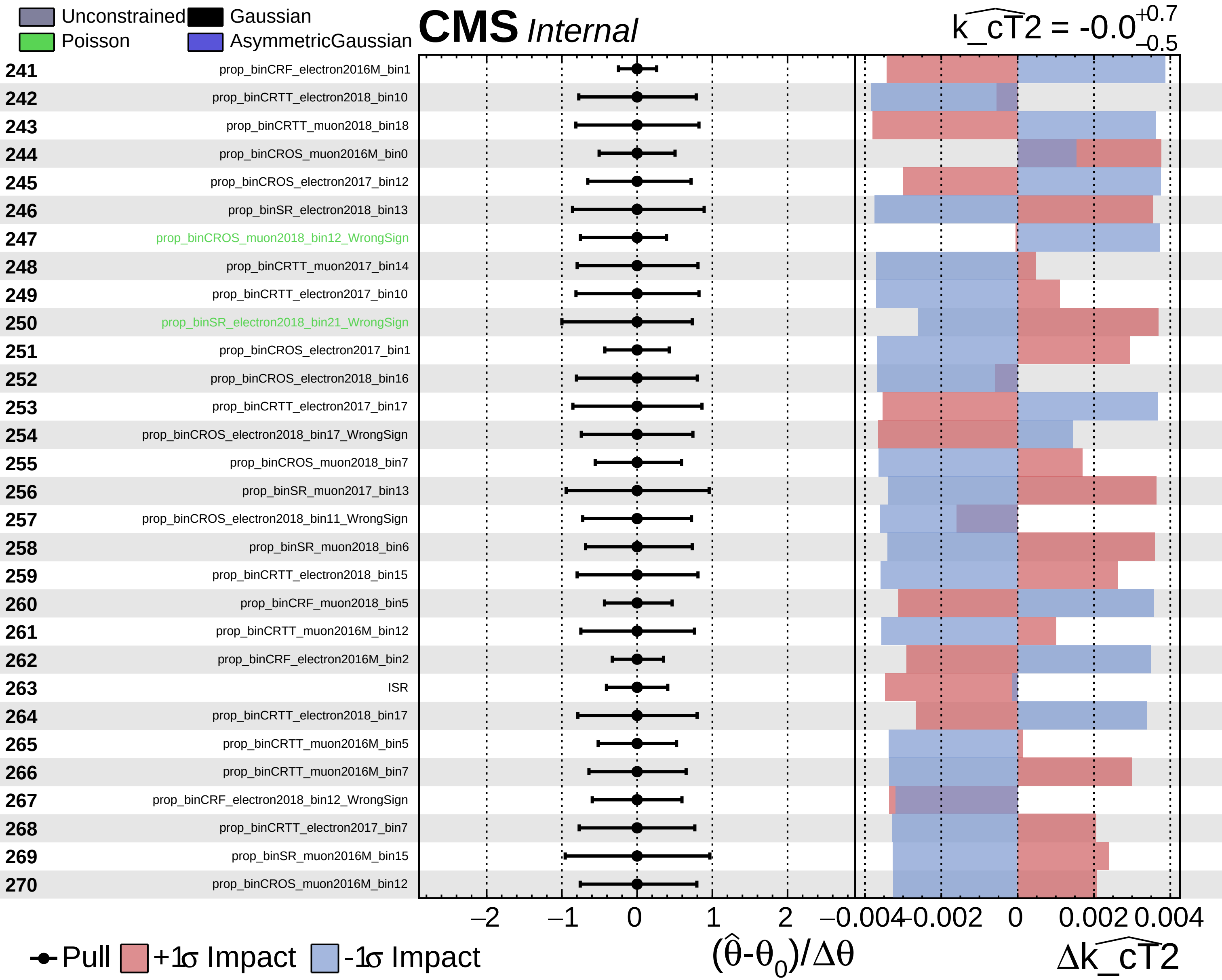
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+0.7}_{-0.5}$
121 prop_binCROS_muon2018_bin11
122 prop_binCRF_muon2016M_bin6
123 FSR
124 lumi_2017
125 prop_binCRTT_electron2018_bin1
126 prop_binSR_muon2017_bin19_sm
127 prop_binCRTT_electron2016M_bin0
128 prop_binSR_muon2018_bin15
129 prop_binSR_muon2018_bin14
130 prop_binCRTT_electron2018_bin2
131 prop_binSR_electron2018_bin21_Triboson
132 TES
133 prop_binCROS_muon2017_bin11_Fake
134 prop_binSR_electron2016M_bin21_TVX
135 QCDScale_Triboson
136 prop_binSR_muon2018_bin10
137 prop_binSR_electron2018_bin4
138 prop_binSR_electron2017_bin16
139 prop_binSR_electron2017_bin20_TTTo2L2Nu
140 prop_binSR_muon2017_bin17
141 prop_binCROS_electron2016M_bin12_Fake
142 prop_binCROS_electron2018_bin8
143 prop_binCROS_electron2016M_bin15
144 prop_binSR_electron2018_bin14
145 prop_binSR_muon2016M_bin19_TTTo2L2Nu
146 prop_binCROS_muon2016M_bin2
147 QCDScale_TVX
148 prop_binCRTT_electron2018_bin0
149 prop_binCROS_muon2017_bin8
150 prop_binCROS_electron2016M_bin4
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
151 prop_binCRTT_muon2018_bin3
152 prop_binCROS_electron2016M_bin7
153 prop_binCROS_electron2018_bin12
154 prop_binSR_muon2018_bin5
155 prop_binCRF_muon2018_bin0
156 prop_binCRF_muon2017_bin1
157 QCDScale_VG
158 prop_binCROS_muon2016M_bin13_WrongSign
159 QCDScale_WrongSign
160 prop_binCRF_muon2018_bin1
161 prop_binCRTT_muon2017_bin2
162 prop_binCRTT_electron2016M_bin4
163 prop_binCRF_electron2018_bin0
164 prop_binCRF_electron2018_bin1
165 prop_binCRTT_electron2016M_bin2
166 prop_binSR_muon2017_bin20_ZZtoLep
167 prop_binCRF_electron2016M_bin16_Fake
168 prop_binCRTT_electron2018_bin14
169 prop_binCROS_electron2017_bin6
170 prop_binCROS_muon2017_bin13_WrongSign
171 prop_binCRTT_muon2017_bin10
172 prop_binCRTT_electron2017_bin3
173 prop_binSR_electron2017_bin20_sm
174 prop_binCROS_muon2018_bin14_Fake
175 prop_binCRF_muon2017_bin11_Fake
176 prop_binCRF_electron2017_bin1
177 prop_binCROS_muon2016M_bin1
178 prop_binCRTT_muon2017_bin5
179 prop_binCRTT_muon2018_bin7
180 prop_binCRTT_electron2018_bin11
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

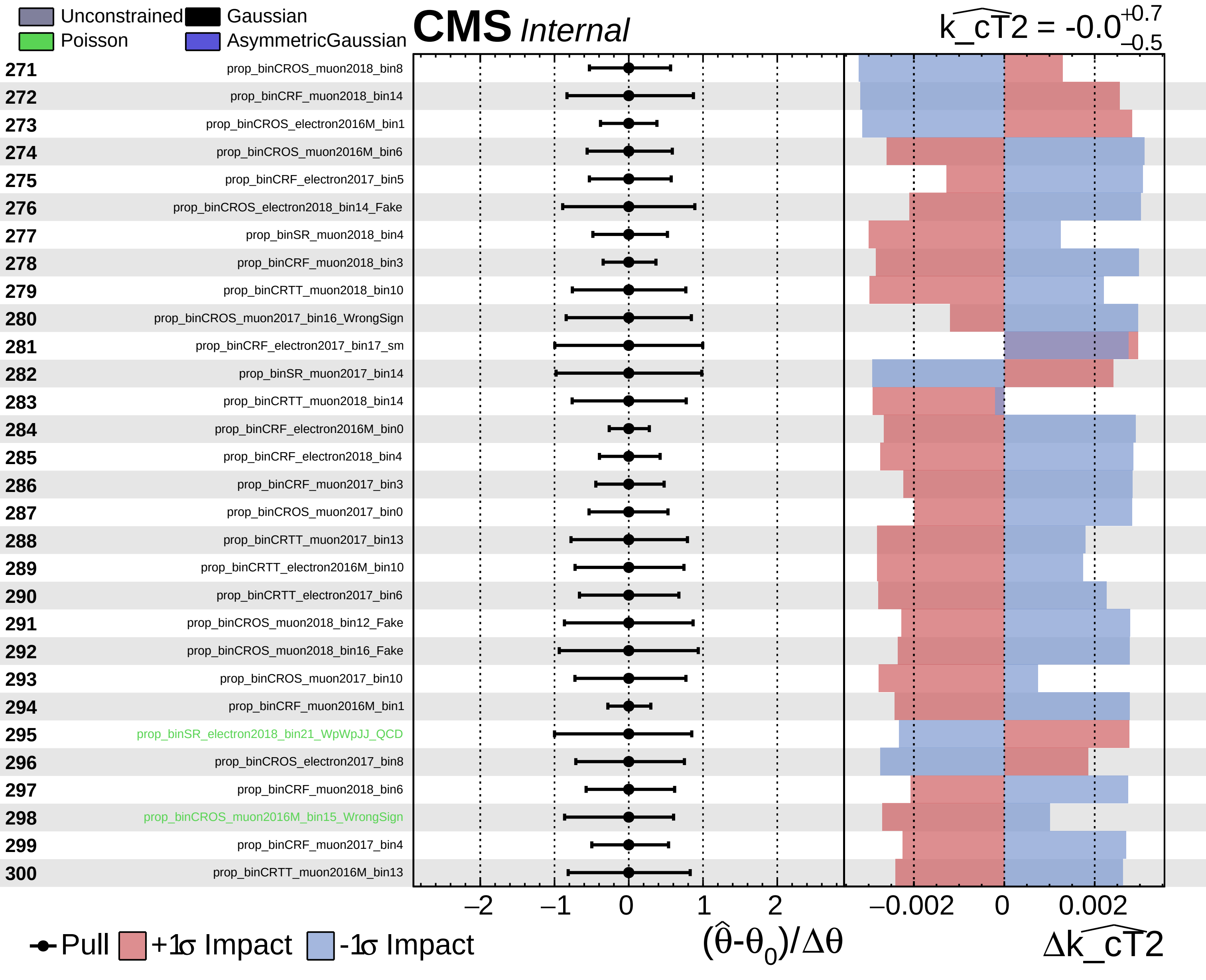


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
181 prop_binCRF_electron2016M_bin3
182 prop_binCRF_electron2018_bin3
183 prop_binCROS_muon2017_bin12_Fake
184 prop_binCRTT_electron2017_bin5
185 prop_binCRTT_electron2018_bin18
186 prop_binCROS_electron2018_bin13_WrongSign
187 prop_binCRF_muon2018_bin20
188 prop_binSR_muon2018_bin13
189 prop_binCRF_electron2016M_bin16_sm
190 prop_binCROS_electron2016M_bin3
191 prop_binCROS_electron2016M_bin16_WrongSign
192 prop_binCRTT_muon2018_bin5
193 prop_binCRF_electron2018_bin2
194 prop_binCROS_muon2018_bin9
195 prop_binCROS_muon2017_bin7
196 prop_binCROS_electron2017_bin4
197 prop_binCROS_electron2018_bin14_WrongSign
198 puID
199 prop_binCRF_muon2017_bin15
200 prop_binCRTT_muon2018_bin9
201 prop_binSR_muon2016M_bin16
202 prop_binCRF_muon2018_bin2
203 prop_binCRF_muon2016M_bin0
204 prop_binSR_electron2017_bin14
205 prop_binCRTT_muon2017_bin17
206 prop_binCRTT_muon2018_bin15
207 prop_binSR_muon2017_bin16
208 prop_binCROS_electron2016M_bin6
209 prop_binCROS_electron2016M_bin14_WrongSign
210 prop_binCRTT_muon2016M_bin2
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

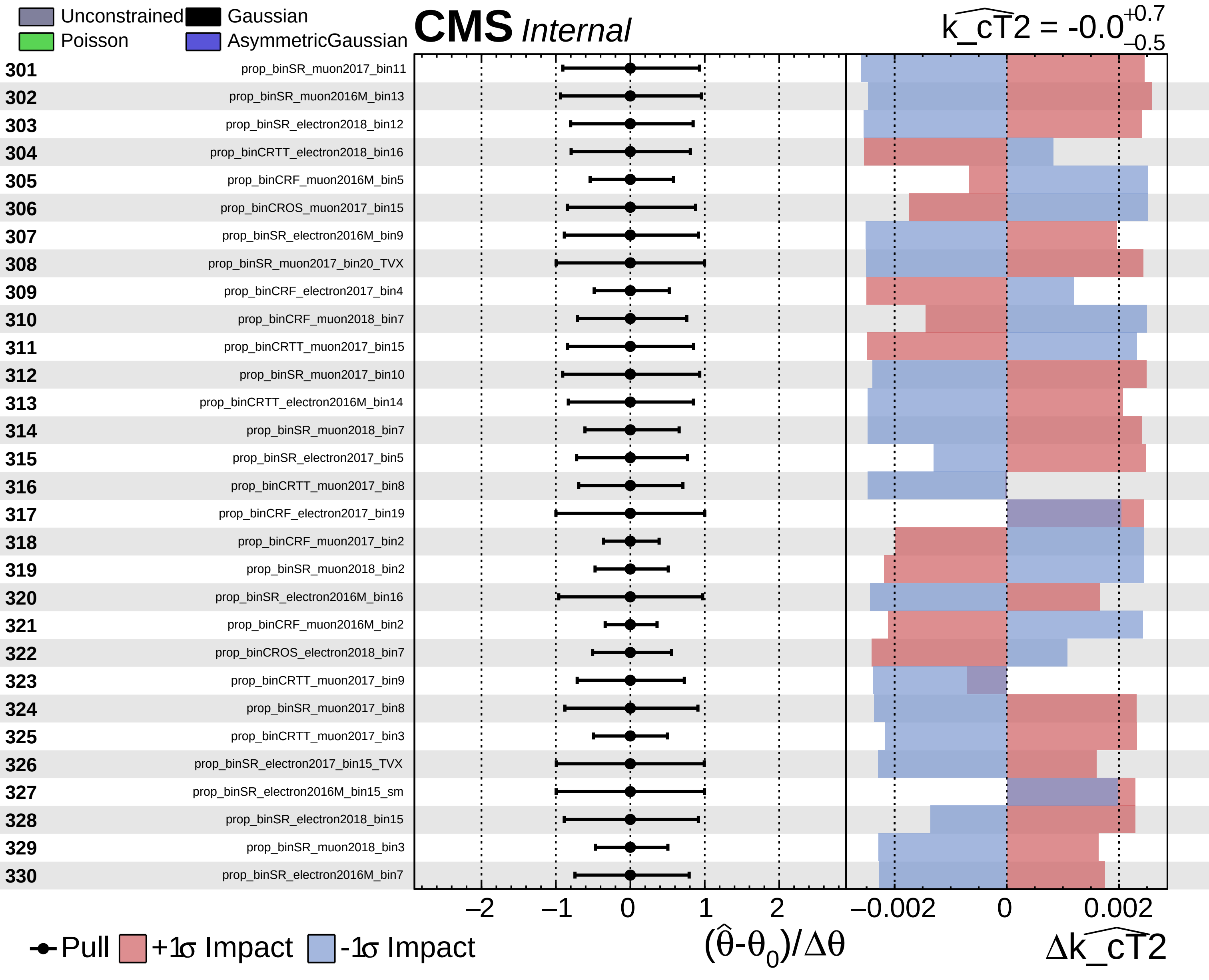

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
211 prop_binCRF_electron2016M_bin10
212 prop_binSR_electron2018_bin8
213 prop_binCRF_muon2017_bin11_sm
214 prop_binCRF_muon2017_bin14_sm
215 prop_binCRF_muon2016M_bin14_sm
216 prop_binCROS_muon2017_bin5
217 prop_binSR_electron2017_bin15_sm
218 prop_binCRTT_electron2018_bin9
219 prop_binCROS_muon2018_bin18
220 prop_binCRF_muon2018_bin4
221 prop_binCROS_muon2016M_bin16
222 prop_binCROS_electron2018_bin3
223 prop_binCRTT_muon2018_bin8
224 prop_binCROS_electron2017_bin7
225 prop_binCRF_muon2016M_bin16_sm
226 prop_binCRTT_electron2016M_bin1
227 prop_binCROS_electron2018_bin9
228 prop_binCRF_electron2017_bin2
229 prop_binCROS_muon2018_bin10
230 prop_binCRF_electron2017_bin18_sm
231 prop_binCRTT_electron2017_bin1
232 prop_binCRTT_electron2018_bin8
233 prop_binCRTT_muon2016M_bin8
234 prop_binSR_muon2016M_bin6
235 prop_binCROS_muon2017_bin17_WrongSign
236 prop_binCRTT_muon2016M_bin0
237 prop_binSR_muon2017_bin12
238 prop_binCRTT_muon2016M_bin10
239 prop_binCROS_electron2016M_bin9
240 prop_binCRF_muon2018_bin18
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



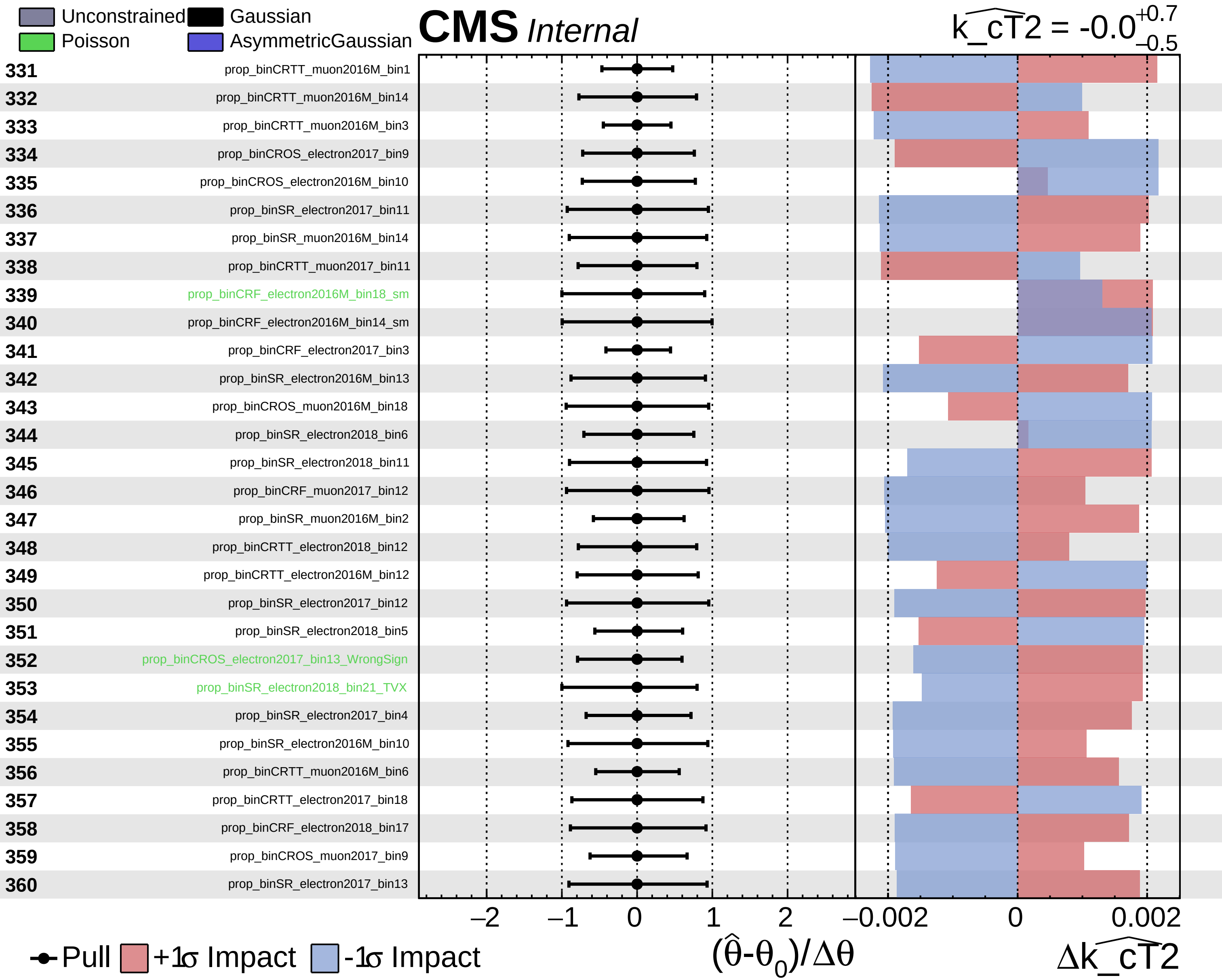
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
241 prop_binCRF_electron2016M_bin1
242 prop_binCRTT_electron2018_bin10
243 prop_binCRTT_muon2018_bin18
244 prop_binCROS_muon2016M_bin0
245 prop_binCROS_electron2017_bin12
246 prop_binSR_electron2018_bin13
247 prop_binCROS_muon2018_bin12_WrongSign
248 prop_binCRTT_muon2017_bin14
249 prop_binCRTT_electron2017_bin10
250 prop_binSR_electron2018_bin21_WrongSign
251 prop_binCROS_electron2017_bin1
252 prop_binCROS_electron2018_bin16
253 prop_binCRTT_electron2017_bin17
254 prop_binCROS_electron2018_bin17_WrongSign
255 prop_binCROS_muon2018_bin7
256 prop_binSR_muon2017_bin13
257 prop_binCROS_electron2018_bin11_WrongSign
258 prop_binSR_muon2018_bin6
259 prop_binCRTT_electron2018_bin15
260 prop_binCRF_muon2018_bin5
261 prop_binCRTT_muon2016M_bin12
262 prop_binCRF_electron2016M_bin2
263 ISR
264 prop_binCRTT_electron2018_bin17
265 prop_binCRTT_muon2016M_bin5
266 prop_binCRTT_muon2016M_bin7
267 prop_binCRF_electron2018_bin12_WrongSign
268 prop_binCRTT_electron2017_bin7
269 prop_binSR_muon2016M_bin15
270 prop_binCROS_muon2016M_bin12
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT2



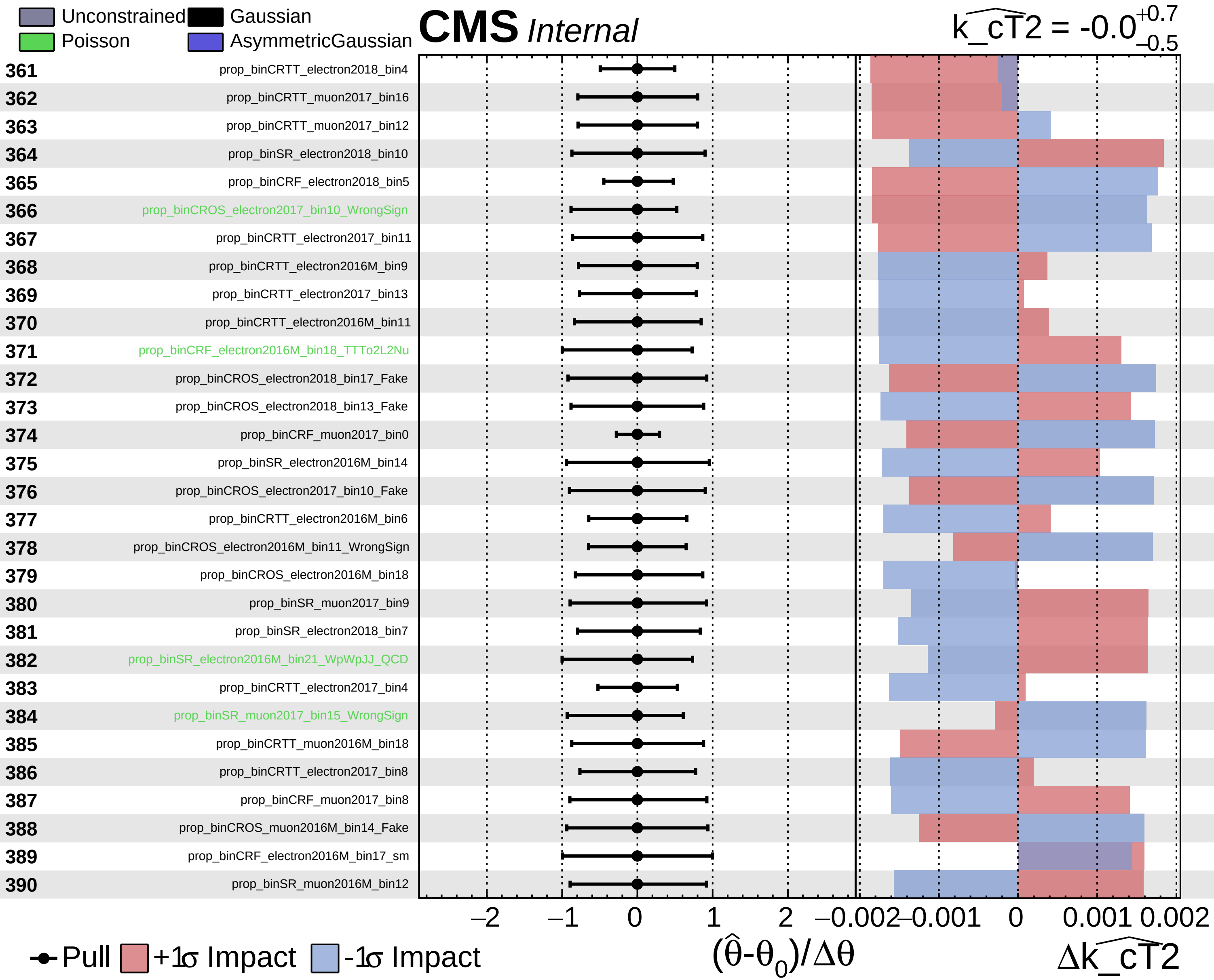
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
271 prop_binCROS_muon2018_bin8
272 prop_binCRF_muon2018_bin14
273 prop_binCROS_electron2016M_bin1
274 prop_binCROS_muon2016M_bin6
275 prop_binCRF_electron2017_bin5
276 prop_binCROS_electron2018_bin14_Fake
277 prop_binSR_muon2018_bin4
278 prop_binCRF_muon2018_bin3
279 prop_binCRTT_muon2018_bin10
280 prop_binCROS_muon2017_bin16_WrongSign
281 prop_binCRF_electron2017_bin17_sm
282 prop_binSR_muon2017_bin14
283 prop_binCRTT_muon2018_bin14
284 prop_binCRF_electron2016M_bin0
285 prop_binCRF_electron2018_bin4
286 prop_binCRF_muon2017_bin3
287 prop_binCROS_muon2017_bin0
288 prop_binCRTT_muon2017_bin13
289 prop_binCRTT_electron2016M_bin10
290 prop_binCRTT_electron2017_bin6
291 prop_binCROS_muon2018_bin12_Fake
292 prop_binCROS_muon2018_bin16_Fake
293 prop_binCROS_muon2017_bin10
294 prop_binCRF_muon2016M_bin1
295 prop_binSR_electron2018_bin21_WpWpJJ_QCD
296 prop_binCROS_electron2017_bin8
297 prop_binCRF_muon2018_bin6
298 prop_binCROS_muon2016M_bin15_WrongSign
299 prop_binCRF_muon2017_bin4
300 prop_binCRTT_muon2016M_bin13
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

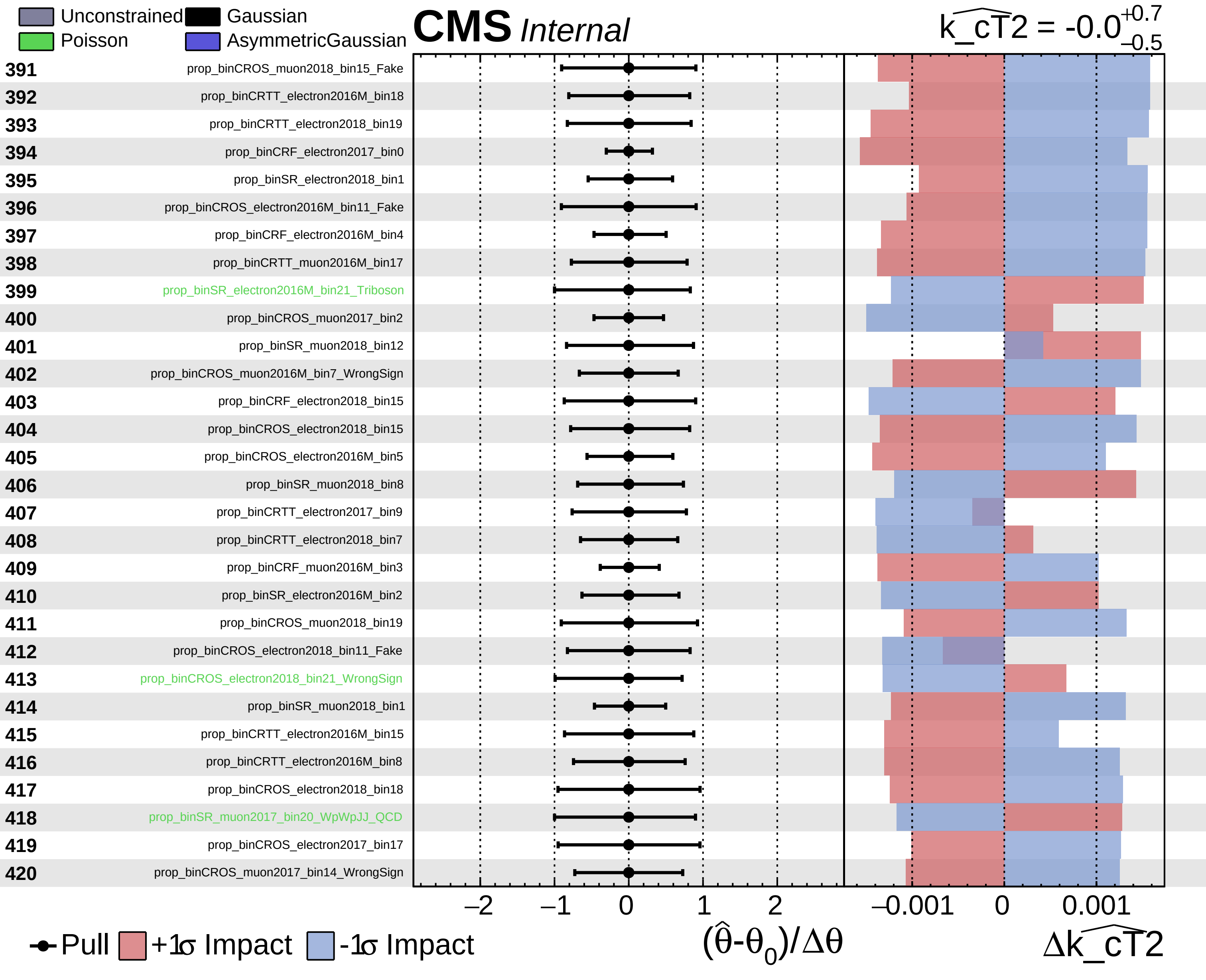


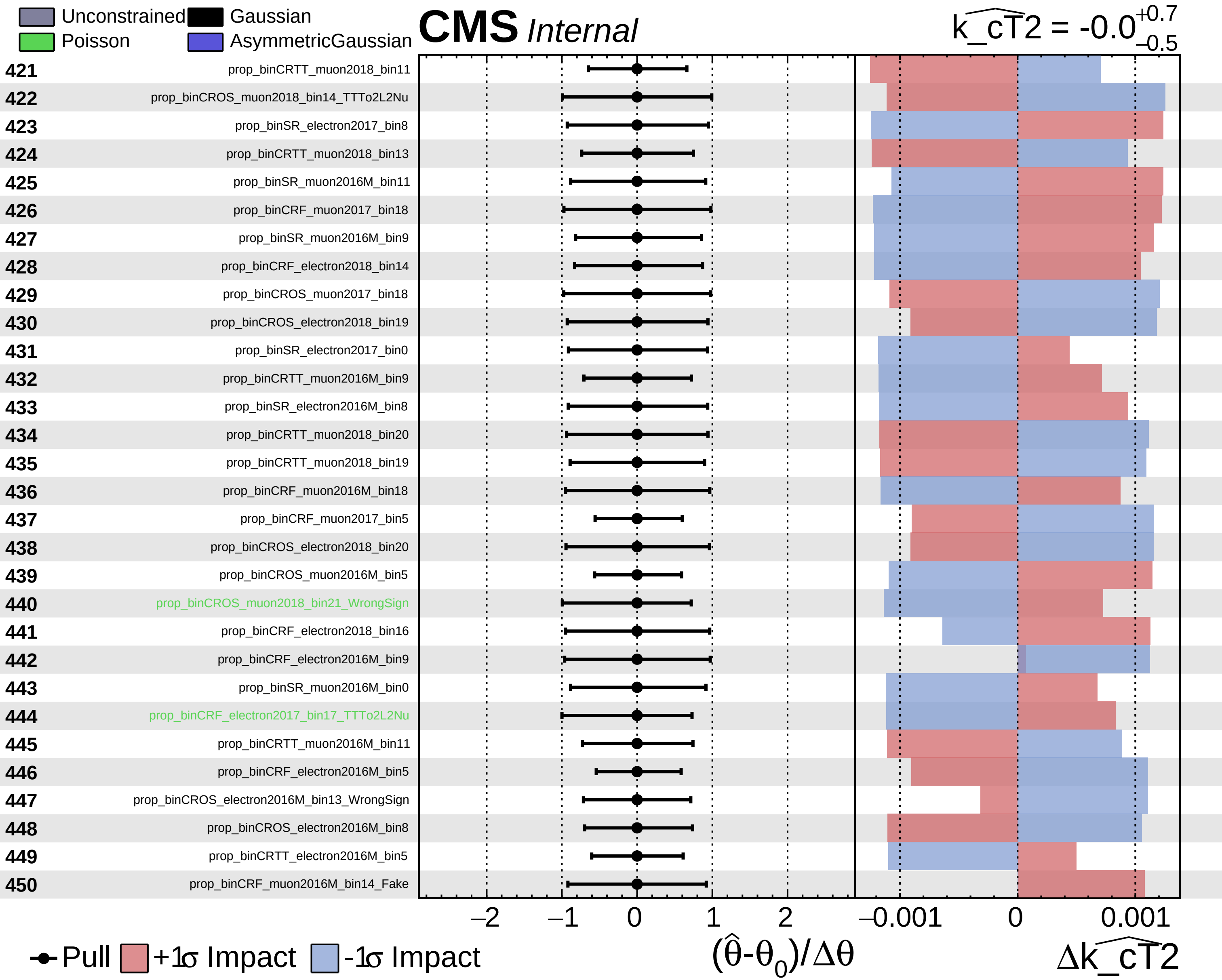
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
301 prop_binSR_muon2017_bin11
302 prop_binSR_muon2016M_bin13
303 prop_binSR_electron2018_bin12
304 prop_binCRTT_electron2018_bin16
305 prop_binCRF_muon2016M_bin5
306 prop_binCROS_muon2017_bin15
307 prop_binSR_electron2016M_bin9
308 prop_binSR_muon2017_bin20_TVX
309 prop_binCRF_electron2017_bin4
310 prop_binCRF_muon2018_bin7
311 prop_binCRTT_muon2017_bin15
312 prop_binSR_muon2017_bin10
313 prop_binCRTT_electron2016M_bin14
314 prop_binSR_muon2018_bin7
315 prop_binSR_electron2017_bin5
316 prop_binCRTT_muon2017_bin8
317 prop_binCRF_electron2017_bin19
318 prop_binCRF_muon2017_bin2
319 prop_binSR_muon2018_bin2
320 prop_binSR_electron2016M_bin16
321 prop_binCRF_muon2016M_bin2
322 prop_binCROS_electron2018_bin7
323 prop_binCRTT_muon2017_bin9
324 prop_binSR_muon2017_bin8
325 prop_binCRTT_muon2017_bin3
326 prop_binSR_electron2017_bin15_TVX
327 prop_binSR_electron2016M_bin15_sm
328 prop_binSR_electron2018_bin15
329 prop_binSR_muon2018_bin3
330 prop_binSR_electron2016M_bin7
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT2



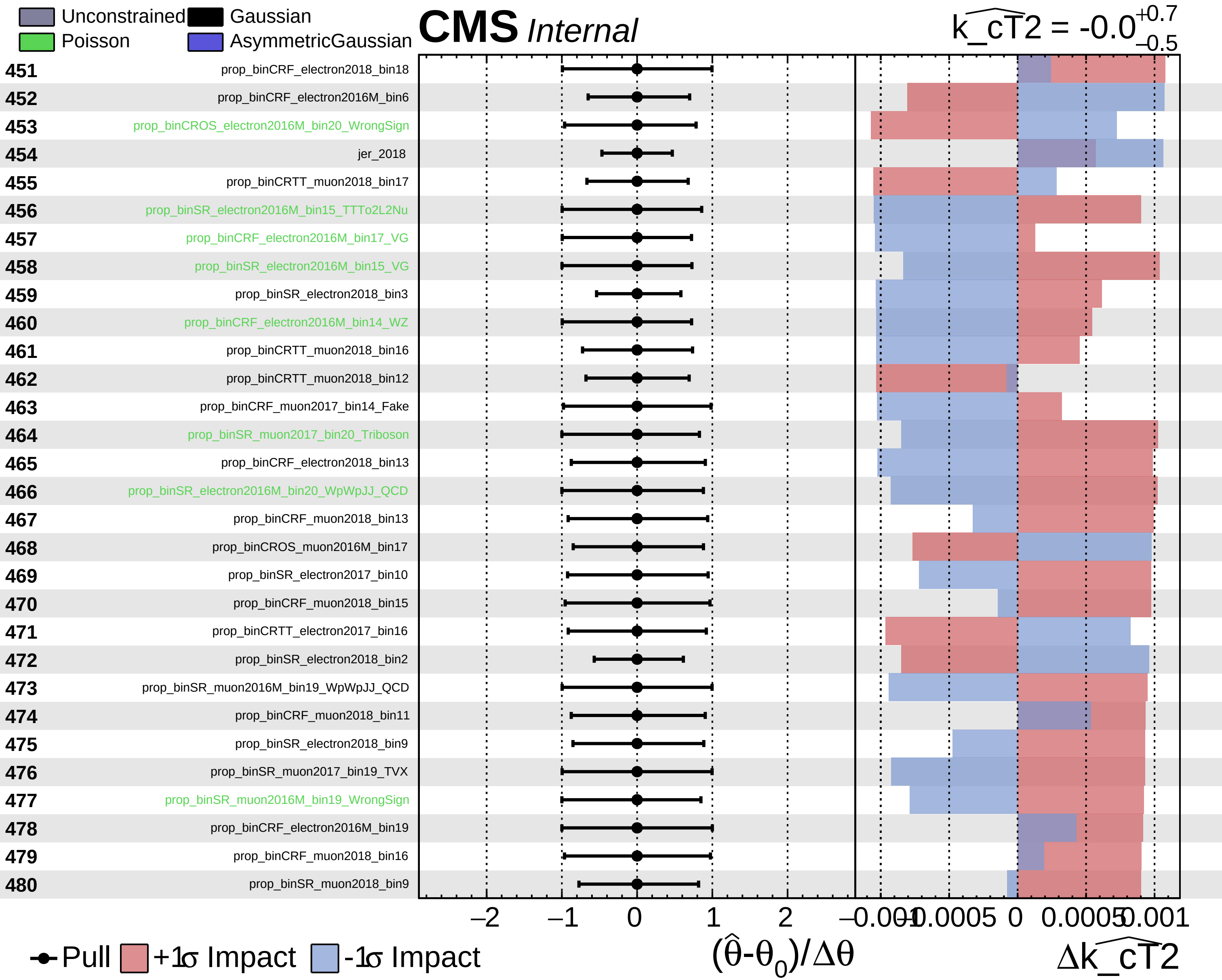
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+0.7}_{-0.5}$
331 prop_binCRTT_muon2016M_bin1
332 prop_binCRTT_muon2016M_bin14
333 prop_binCRTT_muon2016M_bin3
334 prop_binCROS_electron2017_bin9
335 prop_binCROS_electron2016M_bin10
336 prop_binSR_electron2017_bin11
337 prop_binSR_muon2016M_bin14
338 prop_binCRTT_muon2017_bin11
339 prop_binCRF_electron2016M_bin18_sm
340 prop_binCRF_electron2016M_bin14_sm
341 prop_binCRF_electron2017_bin3
342 prop_binSR_electron2016M_bin13
343 prop_binCROS_muon2016M_bin18
344 prop_binSR_electron2018_bin6
345 prop_binSR_electron2018_bin11
346 prop_binCRF_muon2017_bin12
347 prop_binSR_muon2016M_bin2
348 prop_binCRTT_electron2018_bin12
349 prop_binCRTT_electron2016M_bin12
350 prop_binSR_electron2017_bin12
351 prop_binSR_electron2018_bin5
352 prop_binCROS_electron2017_bin13_WrongSign
353 prop_binSR_electron2018_bin21_TVX
354 prop_binSR_electron2017_bin4
355 prop_binSR_electron2016M_bin10
356 prop_binCRTT_muon2016M_bin6
357 prop_binCRTT_electron2017_bin18
358 prop_binCRF_electron2018_bin17
359 prop_binCROS_muon2017_bin9
360 prop_binSR_electron2017_bin13
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
361 prop_binCRTT_electron2018_bin4
362 prop_binCRTT_muon2017_bin16
363 prop_binCRTT_muon2017_bin12
364 prop_binSR_electron2018_bin10
365 prop_binCRF_electron2018_bin5
366 prop_binCROS_electron2017_bin10_WrongSign
367 prop_binCRTT_electron2017_bin11
368 prop_binCRTT_electron2016M_bin9
369 prop_binCRTT_electron2017_bin13
370 prop_binCRTT_electron2016M_bin11
371 prop_binCRF_electron2016M_bin18_TTTo2L2Nu
372 prop_binCROS_electron2018_bin17_Fake
373 prop_binCROS_electron2018_bin13_Fake
374 prop_binCRF_muon2017_bin0
375 prop_binSR_electron2016M_bin14
376 prop_binCROS_electron2017_bin10_Fake
377 prop_binCRTT_electron2016M_bin6
378 prop_binCROS_electron2016M_bin11_WrongSign
379 prop_binCROS_electron2016M_bin18
380 prop_binSR_muon2017_bin9
381 prop_binSR_electron2018_bin7
382 prop_binSR_electron2016M_bin21_WpWpJJ_QCD
383 prop_binCRTT_electron2017_bin4
384 prop_binSR_muon2017_bin15_WrongSign
385 prop_binCRTT_muon2016M_bin18
386 prop_binCRTT_electron2017_bin8
387 prop_binCRF_muon2017_bin8
388 prop_binCROS_muon2016M_bin14_Fake
389 prop_binCRF_electron2016M_bin17_sm
390 prop_binSR_muon2016M_bin12
−2
−1
0
1
2
−0.002
−0.001
0
0.001
0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

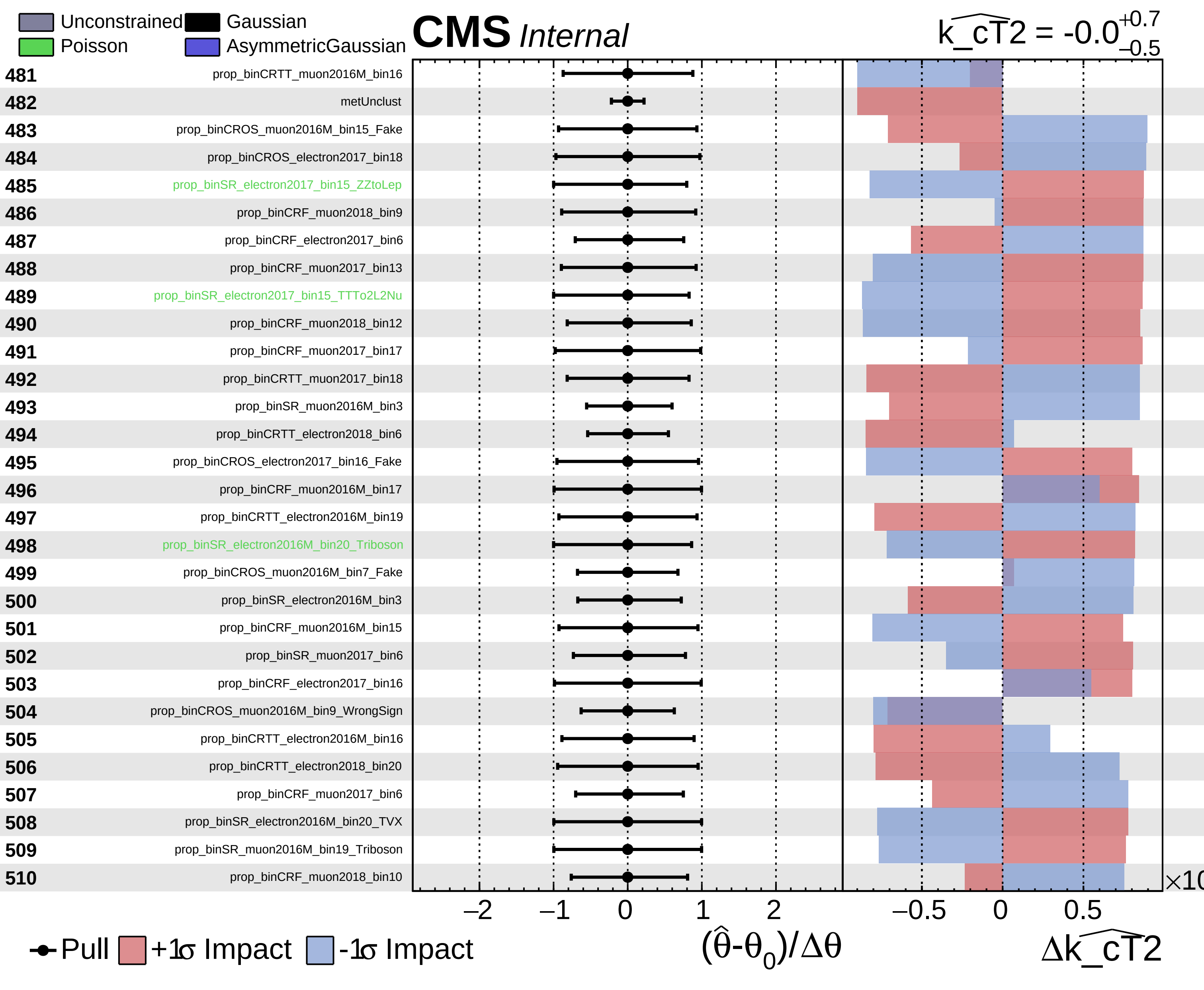

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
391 prop_binCROS_muon2018_bin15_Fake
392 prop_binCRTT_electron2016M_bin18
393 prop_binCRTT_electron2018_bin19
394 prop_binCRF_electron2017_bin0
395 prop_binSR_electron2018_bin1
396 prop_binCROS_electron2016M_bin11_Fake
397 prop_binCRF_electron2016M_bin4
398 prop_binCRTT_muon2016M_bin17
399 prop_binSR_electron2016M_bin21_Triboson
400 prop_binCROS_muon2017_bin2
401 prop_binSR_muon2018_bin12
402 prop_binCROS_muon2016M_bin7_WrongSign
403 prop_binCRF_electron2018_bin15
404 prop_binCROS_electron2018_bin15
405 prop_binCROS_electron2016M_bin5
406 prop_binSR_muon2018_bin8
407 prop_binCRTT_electron2017_bin9
408 prop_binCRTT_electron2018_bin7
409 prop_binCRF_muon2016M_bin3
410 prop_binSR_electron2016M_bin2
411 prop_binCROS_muon2018_bin19
412 prop_binCROS_electron2018_bin11_Fake
413 prop_binCROS_electron2018_bin21_WrongSign
414 prop_binSR_muon2018_bin1
415 prop_binCRTT_electron2016M_bin15
416 prop_binCRTT_electron2016M_bin8
417 prop_binCROS_electron2018_bin18
418 prop_binSR_muon2017_bin20_WpWpJJ_QCD
419 prop_binCROS_electron2017_bin17
420 prop_binCROS_muon2017_bin14_WrongSign
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

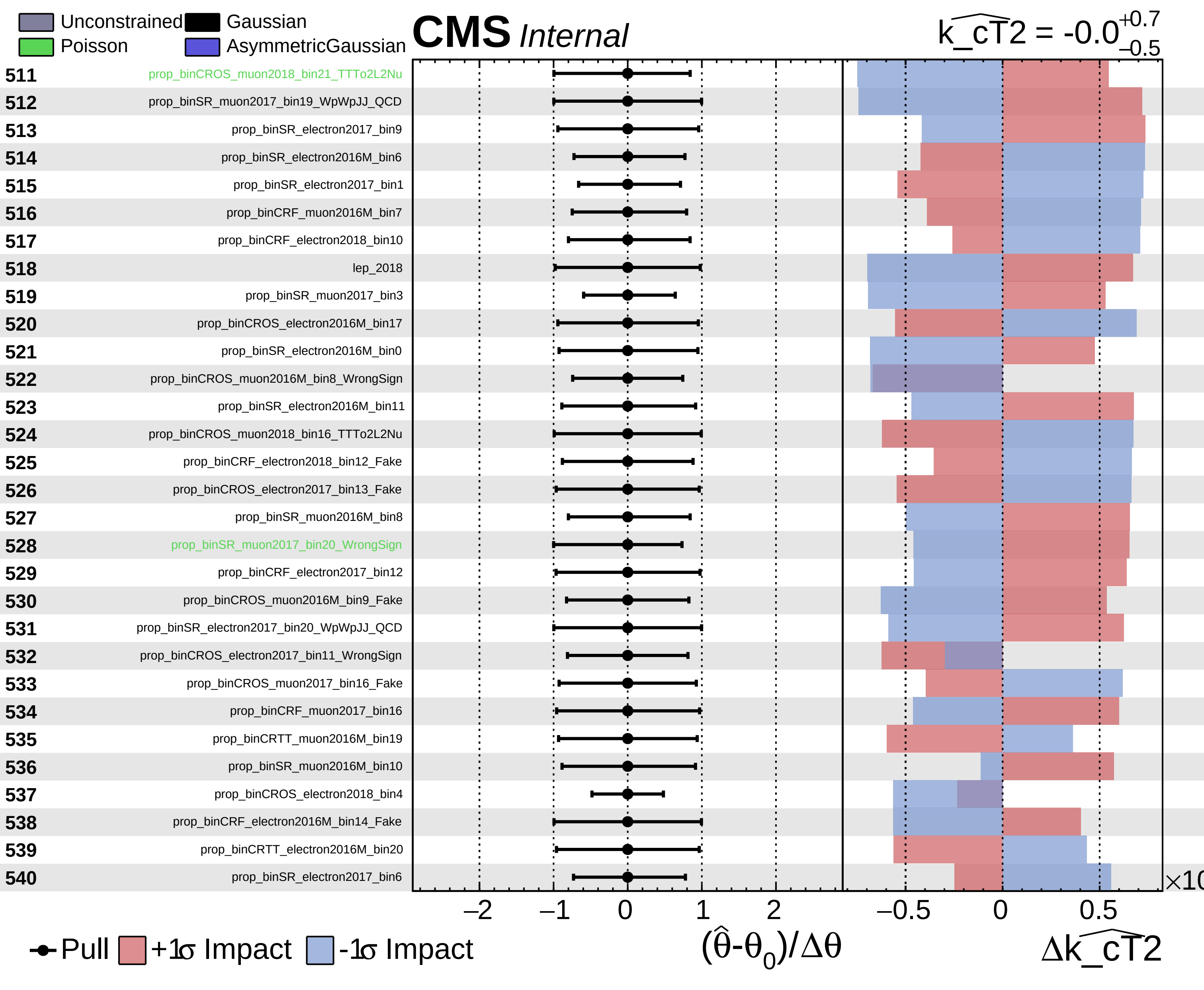


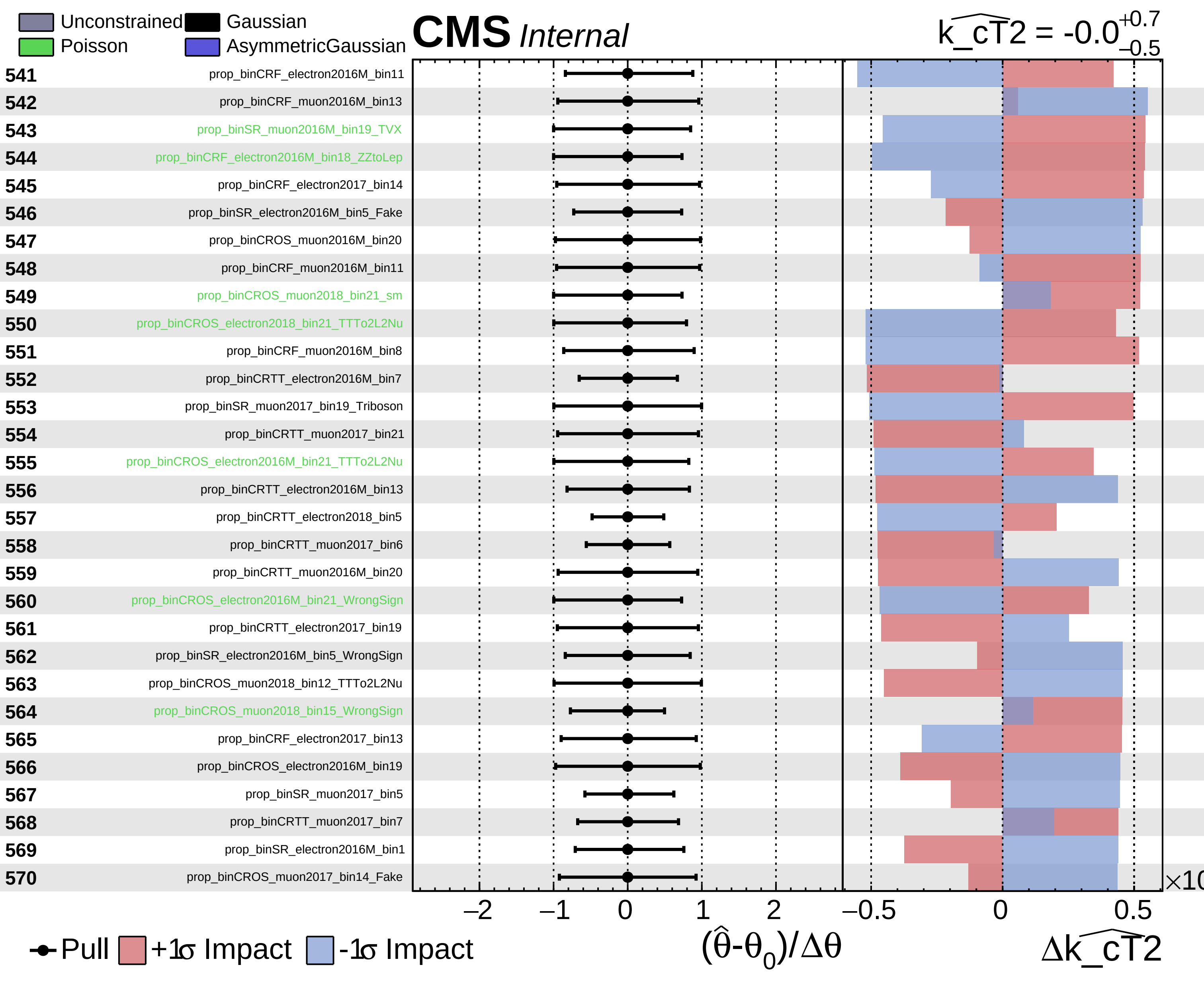
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
421 prop_binCRTT_muon2018_bin11
422 prop_binCROS_muon2018_bin14_TTTo2L2Nu
423 prop_binSR_electron2017_bin8
424 prop_binCRTT_muon2018_bin13
425 prop_binSR_muon2016M_bin11
426 prop_binCRF_muon2017_bin18
427 prop_binSR_muon2016M_bin9
428 prop_binCRF_electron2018_bin14
429 prop_binCROS_muon2017_bin18
430 prop_binCROS_electron2018_bin19
431 prop_binSR_electron2017_bin0
432 prop_binCRTT_muon2016M_bin9
433 prop_binSR_electron2016M_bin8
434 prop_binCRTT_muon2018_bin20
435 prop_binCRTT_muon2018_bin19
436 prop_binCRF_muon2016M_bin18
437 prop_binCRF_muon2017_bin5
438 prop_binCROS_electron2018_bin20
439 prop_binCROS_muon2016M_bin5
440 prop_binCROS_muon2018_bin21_WrongSign
441 prop_binCRF_electron2018_bin16
442 prop_binCRF_electron2016M_bin9
443 prop_binSR_muon2016M_bin0
444 prop_binCRF_electron2017_bin17_TTTo2L2Nu
445 prop_binCRTT_muon2016M_bin11
446 prop_binCRF_electron2016M_bin5
447 prop_binCROS_electron2016M_bin13_WrongSign
448 prop_binCROS_electron2016M_bin8
449 prop_binCRTT_electron2016M_bin5
450 prop_binCRF_muon2016M_bin14_Fake
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



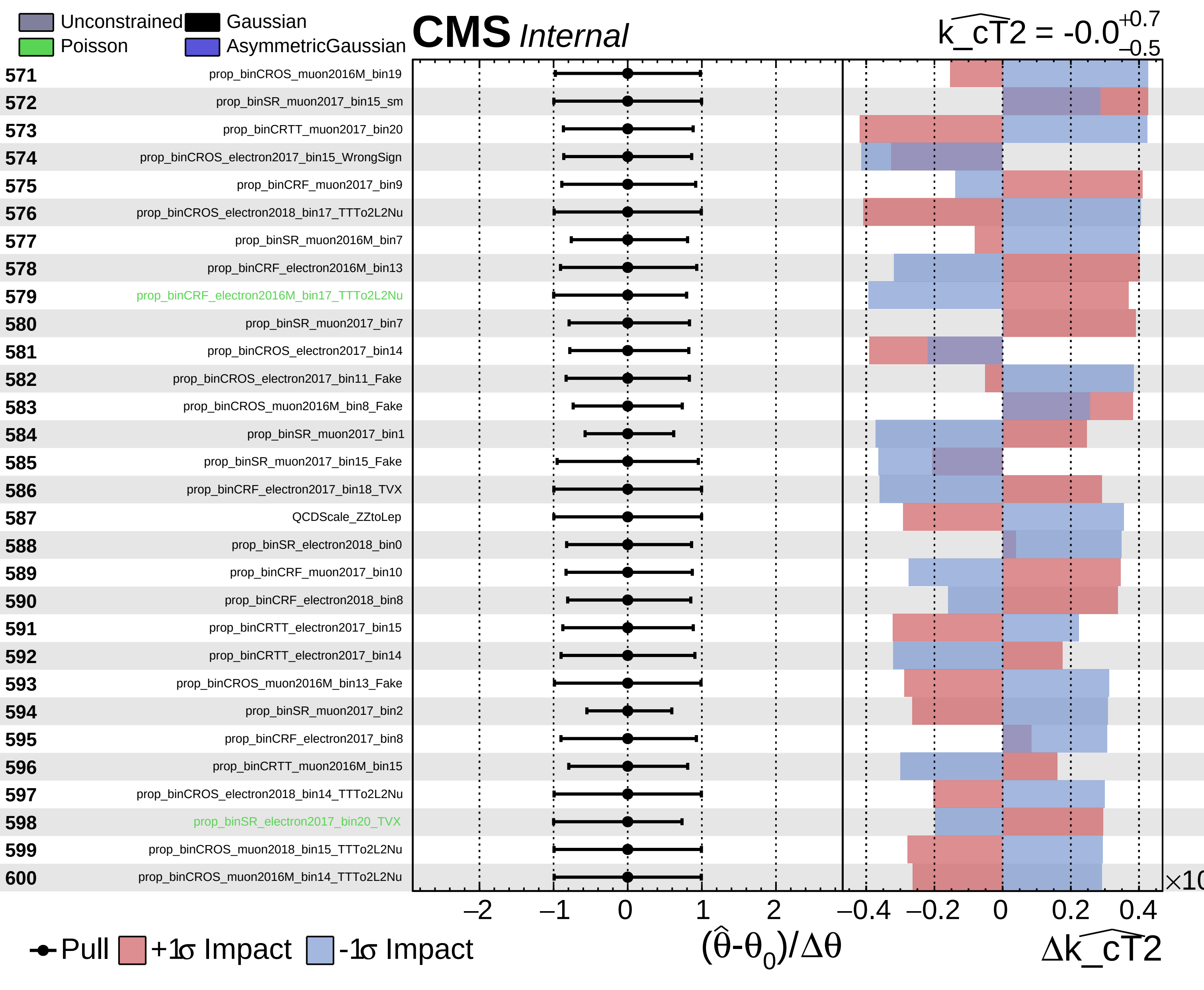
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
451 prop_binCRF_electron2018_bin18
452 prop_binCRF_electron2016M_bin6
453 prop_binCROS_electron2016M_bin20_WrongSign
454 jer_2018
455 prop_binCRTT_muon2018_bin17
456 prop_binSR_electron2016M_bin15_TTTo2L2Nu
457 prop_binCRF_electron2016M_bin17_VG
458 prop_binSR_electron2016M_bin15_VG
459 prop_binSR_electron2018_bin3
460 prop_binCRF_electron2016M_bin14_WZ
461 prop_binCRTT_muon2018_bin16
462 prop_binCRTT_muon2018_bin12
463 prop_binCRF_muon2017_bin14_Fake
464 prop_binSR_muon2017_bin20_Triboson
465 prop_binCRF_electron2018_bin13
466 prop_binSR_electron2016M_bin20_WpWpJJ_QCD
467 prop_binCRF_muon2018_bin13
468 prop_binCROS_muon2016M_bin17
469 prop_binSR_electron2017_bin10
470 prop_binCRF_muon2018_bin15
471 prop_binCRTT_electron2017_bin16
472 prop_binSR_electron2018_bin2
473 prop_binSR_muon2016M_bin19_WpWpJJ_QCD
474 prop_binCRF_muon2018_bin11
475 prop_binSR_electron2018_bin9
476 prop_binSR_muon2017_bin19_TVX
477 prop_binSR_muon2016M_bin19_WrongSign
478 prop_binCRF_electron2016M_bin19
479 prop_binCRF_muon2018_bin16
480 prop_binSR_muon2018_bin9
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT2


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
481 prop_binCRTT_muon2016M_bin16
482 metUnclust
483 prop_binCROS_muon2016M_bin15_Fake
484 prop_binCROS_electron2017_bin18
485 prop_binSR_electron2017_bin15_ZZtoLep
486 prop_binCRF_muon2018_bin9
487 prop_binCRF_electron2017_bin6
488 prop_binCRF_muon2017_bin13
489 prop_binSR_electron2017_bin15_TTTo2L2Nu
490 prop_binCRF_muon2018_bin12
491 prop_binCRF_muon2017_bin17
492 prop_binCRTT_muon2017_bin18
493 prop_binSR_muon2016M_bin3
494 prop_binCRTT_electron2018_bin6
495 prop_binCROS_electron2017_bin16_Fake
496 prop_binCRF_muon2016M_bin17
497 prop_binCRTT_electron2016M_bin19
498 prop_binSR_electron2016M_bin20_Triboson
499 prop_binCROS_muon2016M_bin7_Fake
500 prop_binSR_electron2016M_bin3
501 prop_binCRF_muon2016M_bin15
502 prop_binSR_muon2017_bin6
503 prop_binCRF_electron2017_bin16
504 prop_binCROS_muon2016M_bin9_WrongSign
505 prop_binCRTT_electron2016M_bin16
506 prop_binCRTT_electron2018_bin20
507 prop_binCRF_muon2017_bin6
508 prop_binSR_electron2016M_bin20_TVX
509 prop_binSR_muon2016M_bin19_Triboson
510 prop_binCRF_muon2018_bin10
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

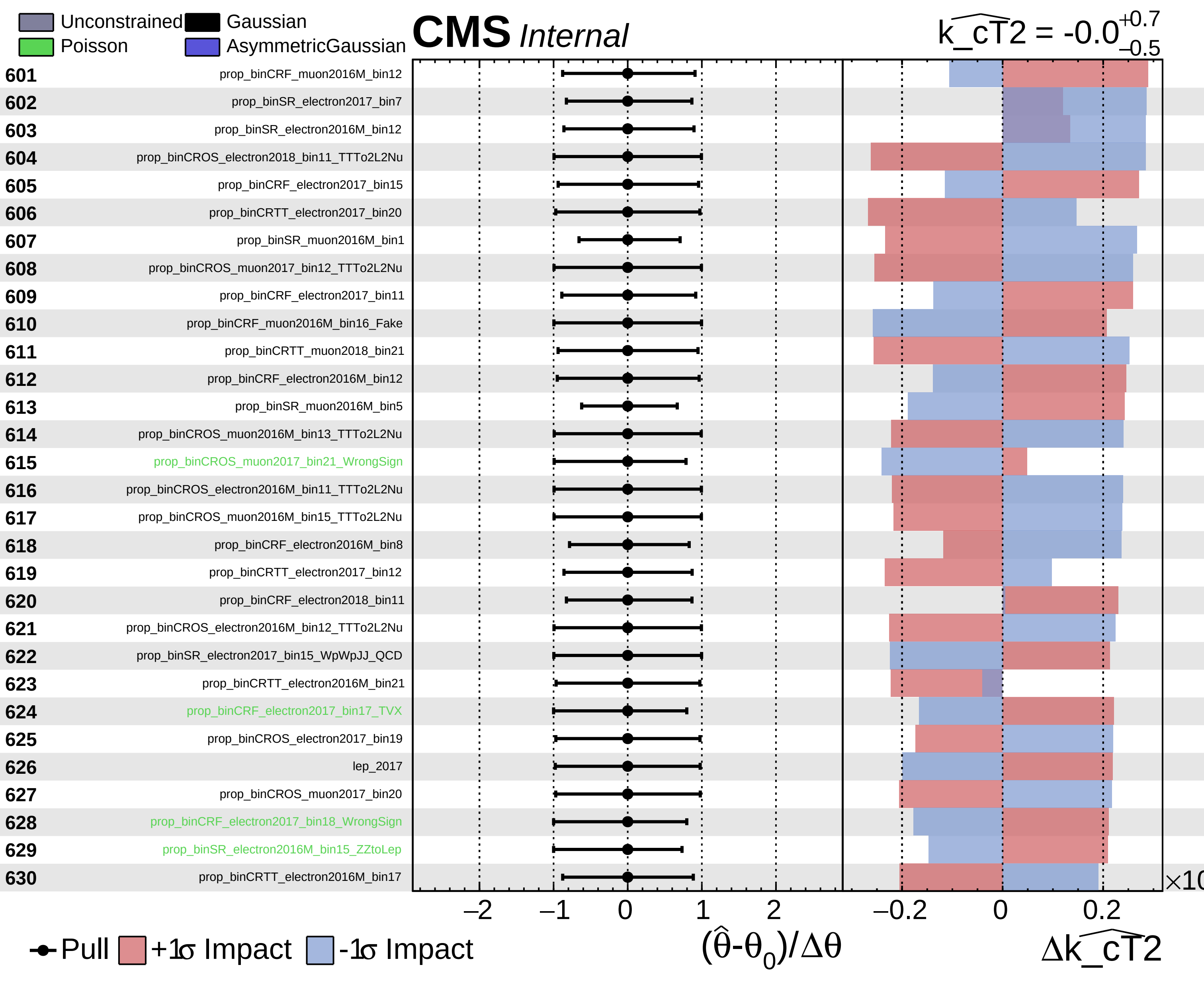

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0+0.7−0.5
511 prop_binCROS_muon2018_bin21_TTTo2L2Nu
512 prop_binSR_muon2017_bin19_WpWpJJ_QCD
513 prop_binSR_electron2017_bin9
514 prop_binSR_electron2016M_bin6
515 prop_binSR_electron2017_bin1
516 prop_binCRF_muon2016M_bin7
517 prop_binCRF_electron2018_bin10
518 lep_2018
519 prop_binSR_muon2017_bin3
520 prop_binCROS_electron2016M_bin17
521 prop_binSR_electron2016M_bin0
522 prop_binCROS_muon2016M_bin8_WrongSign
523 prop_binSR_electron2016M_bin11
524 prop_binCROS_muon2018_bin16_TTTo2L2Nu
525 prop_binCRF_electron2018_bin12_Fake
526 prop_binCROS_electron2017_bin13_Fake
527 prop_binSR_muon2016M_bin8
528 prop_binSR_muon2017_bin20_WrongSign
529 prop_binCRF_electron2017_bin12
530 prop_binCROS_muon2016M_bin9_Fake
531 prop_binSR_electron2017_bin20_WpWpJJ_QCD
532 prop_binCROS_electron2017_bin11_WrongSign
533 prop_binCROS_muon2017_bin16_Fake
534 prop_binCRF_muon2017_bin16
535 prop_binCRTT_muon2016M_bin19
536 prop_binSR_muon2016M_bin10
537 prop_binCROS_electron2018_bin4
538 prop_binCRF_electron2016M_bin14_Fake
539 prop_binCRTT_electron2016M_bin20
540 prop_binSR_electron2017_bin6
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



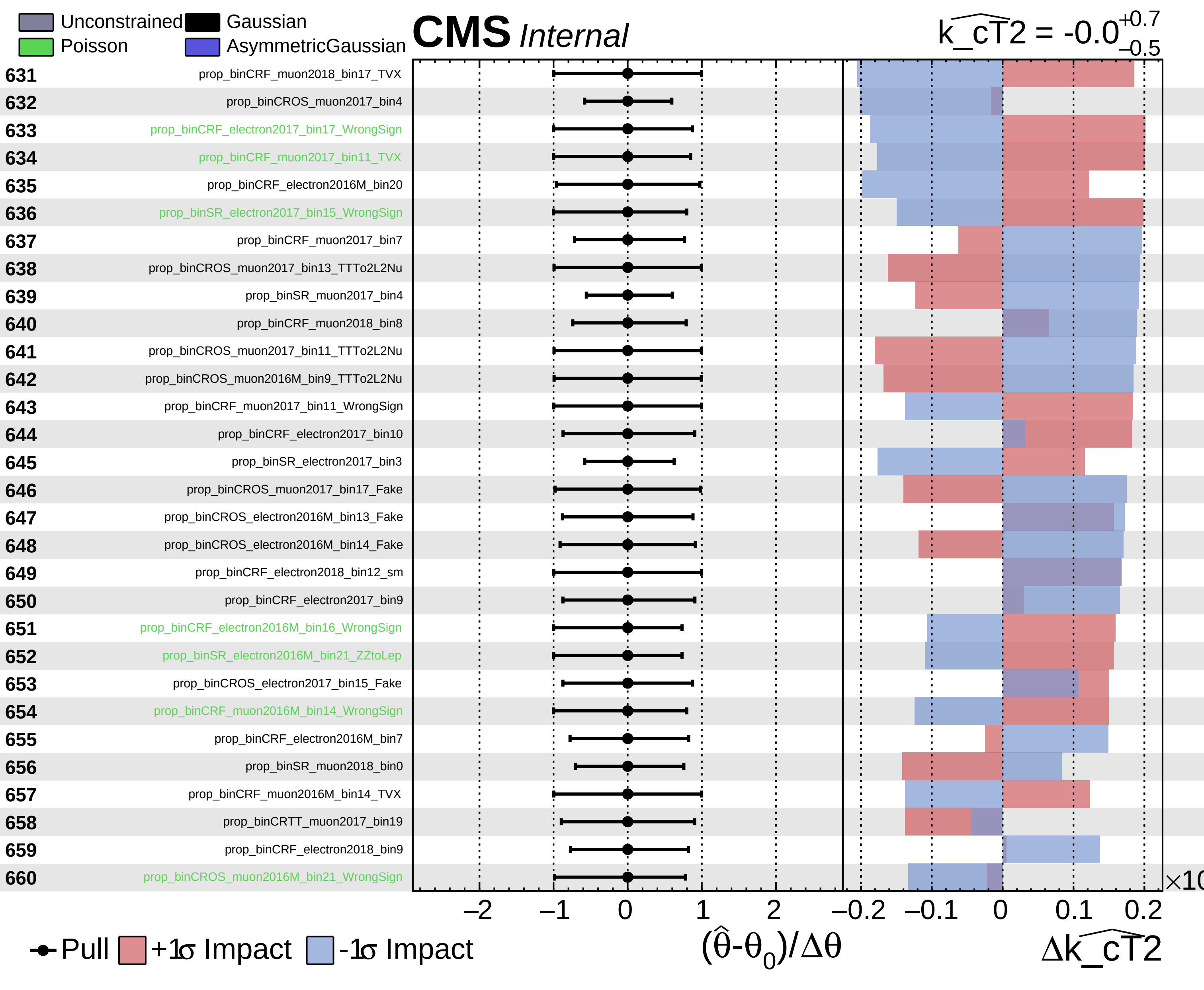
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
541 prop_binCRF_electron2016M_bin11
542 prop_binCRF_muon2016M_bin13
543 prop_binSR_muon2016M_bin19_TVX
544 prop_binCRF_electron2016M_bin18_ZZtoLep
545 prop_binCRF_electron2017_bin14
546 prop_binSR_electron2016M_bin5_Fake
547 prop_binCROS_muon2016M_bin20
548 prop_binCRF_muon2016M_bin11
549 prop_binCROS_muon2018_bin21_sm
550 prop_binCROS_electron2018_bin21_TTTo2L2Nu
551 prop_binCRF_muon2016M_bin8
552 prop_binCRTT_electron2016M_bin7
553 prop_binSR_muon2017_bin19_Triboson
554 prop_binCRTT_muon2017_bin21
555 prop_binCROS_electron2016M_bin21_TTTo2L2Nu
556 prop_binCRTT_electron2016M_bin13
557 prop_binCRTT_electron2018_bin5
558 prop_binCRTT_muon2017_bin6
559 prop_binCRTT_muon2016M_bin20
560 prop_binCROS_electron2016M_bin21_WrongSign
561 prop_binCRTT_electron2017_bin19
562 prop_binSR_electron2016M_bin5_WrongSign
563 prop_binCROS_muon2018_bin12_TTTo2L2Nu
564 prop_binCROS_muon2018_bin15_WrongSign
565 prop_binCRF_electron2017_bin13
566 prop_binCROS_electron2016M_bin19
567 prop_binSR_muon2017_bin5
568 prop_binCRTT_muon2017_bin7
569 prop_binSR_electron2016M_bin1
570 prop_binCROS_muon2017_bin14_Fake
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



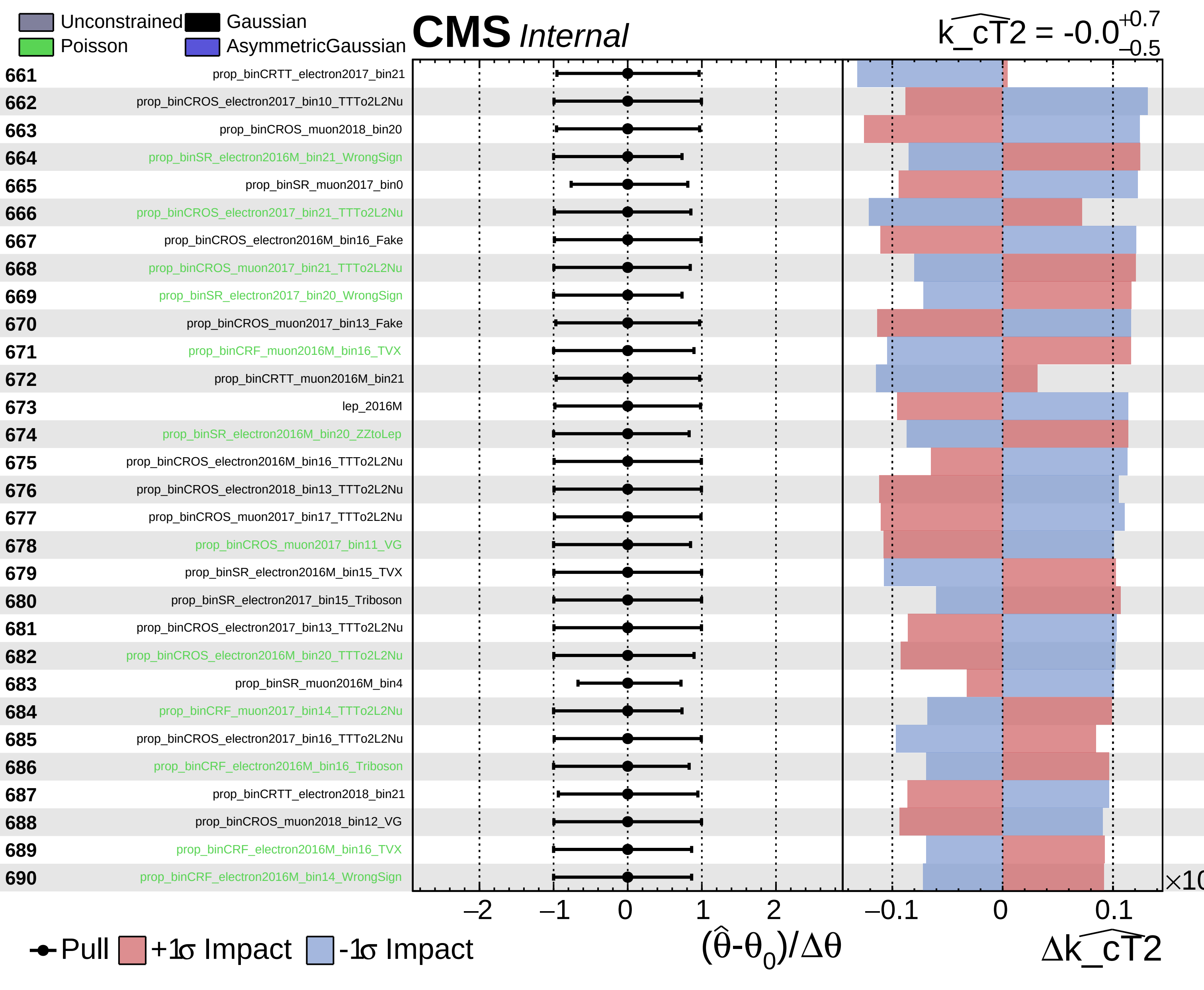
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
571 prop_binCROS_muon2016M_bin19
572 prop_binSR_muon2017_bin15_sm
573 prop_binCRTT_muon2017_bin20
574 prop_binCROS_electron2017_bin15_WrongSign
575 prop_binCRF_muon2017_bin9
576 prop_binCROS_electron2018_bin17_TTTo2L2Nu
577 prop_binSR_muon2016M_bin7
578 prop_binCRF_electron2016M_bin13
579 prop_binCRF_electron2016M_bin17_TTTo2L2Nu
580 prop_binSR_muon2017_bin7
581 prop_binCROS_electron2017_bin14
582 prop_binCROS_electron2017_bin11_Fake
583 prop_binCROS_muon2016M_bin8_Fake
584 prop_binSR_muon2017_bin1
585 prop_binSR_muon2017_bin15_Fake
586 prop_binCRF_electron2017_bin18_TVX
587 QCDScale_ZZtoLep
588 prop_binSR_electron2018_bin0
589 prop_binCRF_muon2017_bin10
590 prop_binCRF_electron2018_bin8
591 prop_binCRTT_electron2017_bin15
592 prop_binCRTT_electron2017_bin14
593 prop_binCROS_muon2016M_bin13_Fake
594 prop_binSR_muon2017_bin2
595 prop_binCRF_electron2017_bin8
596 prop_binCRTT_muon2016M_bin15
597 prop_binCROS_electron2018_bin14_TTTo2L2Nu
598 prop_binSR_electron2017_bin20_TVX
599 prop_binCROS_muon2018_bin15_TTTo2L2Nu
600 prop_binCROS_muon2016M_bin14_TTTo2L2Nu
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

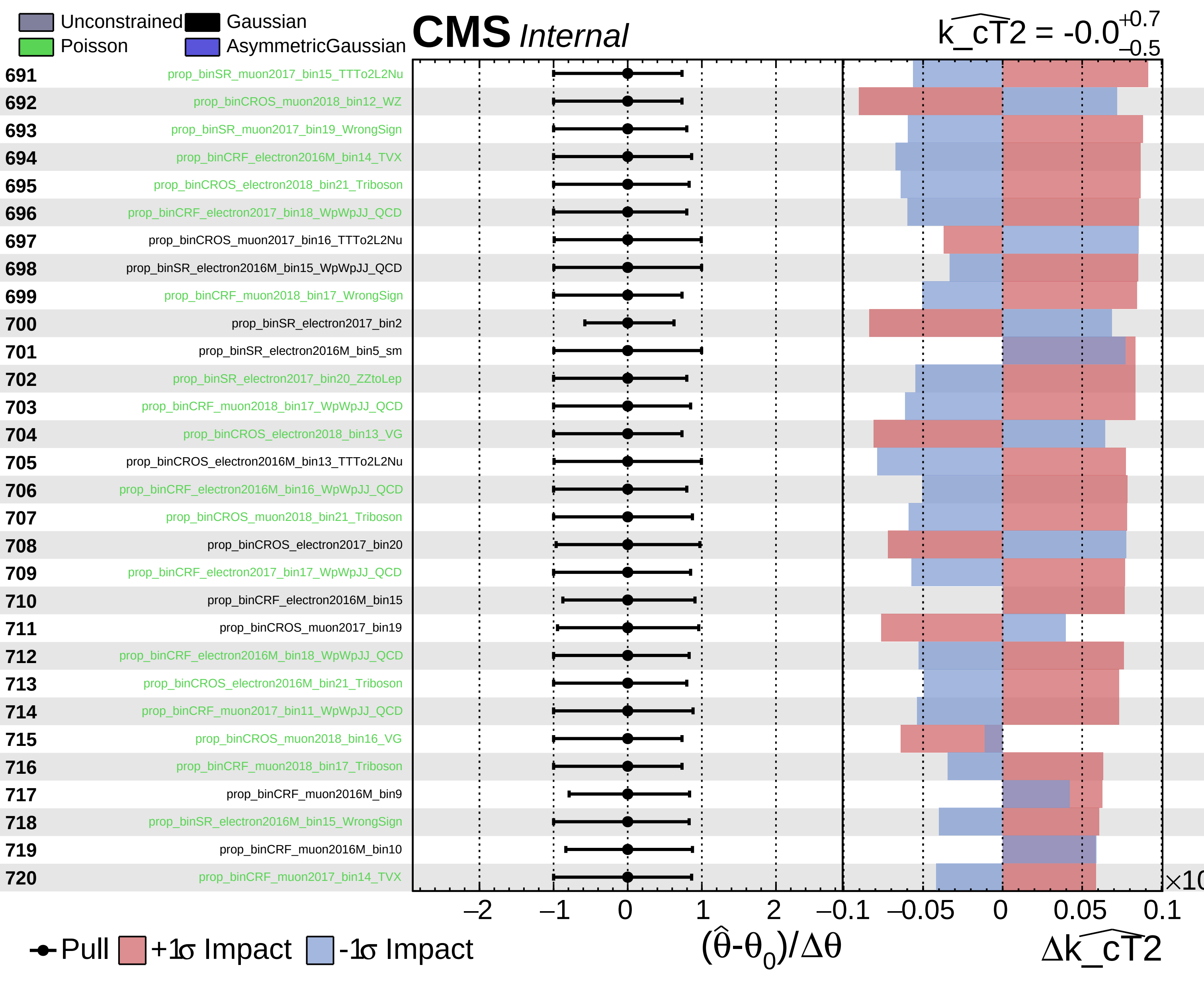


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
601 prop_binCRF_muon2016M_bin12
602 prop_binSR_electron2017_bin7
603 prop_binSR_electron2016M_bin12
604 prop_binCROS_electron2018_bin11_TTTo2L2Nu
605 prop_binCRF_electron2017_bin15
606 prop_binCRTT_electron2017_bin20
607 prop_binSR_muon2016M_bin1
608 prop_binCROS_muon2017_bin12_TTTo2L2Nu
609 prop_binCRF_electron2017_bin11
610 prop_binCRF_muon2016M_bin16_Fake
611 prop_binCRTT_muon2018_bin21
612 prop_binCRF_electron2016M_bin12
613 prop_binSR_muon2016M_bin5
614 prop_binCROS_muon2016M_bin13_TTTo2L2Nu
615 prop_binCROS_muon2017_bin21_WrongSign
616 prop_binCROS_electron2016M_bin11_TTTo2L2Nu
617 prop_binCROS_muon2016M_bin15_TTTo2L2Nu
618 prop_binCRF_electron2016M_bin8
619 prop_binCRTT_electron2017_bin12
620 prop_binCRF_electron2018_bin11
621 prop_binCROS_electron2016M_bin12_TTTo2L2Nu
622 prop_binSR_electron2017_bin15_WpWpJJ_QCD
623 prop_binCRTT_electron2016M_bin21
624 prop_binCRF_electron2017_bin17_TVX
625 prop_binCROS_electron2017_bin19
626 lep_2017
627 prop_binCROS_muon2017_bin20
628 prop_binCRF_electron2017_bin18_WrongSign
629 prop_binSR_electron2016M_bin15_ZZtoLep
630 prop_binCRTT_electron2016M_bin17
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

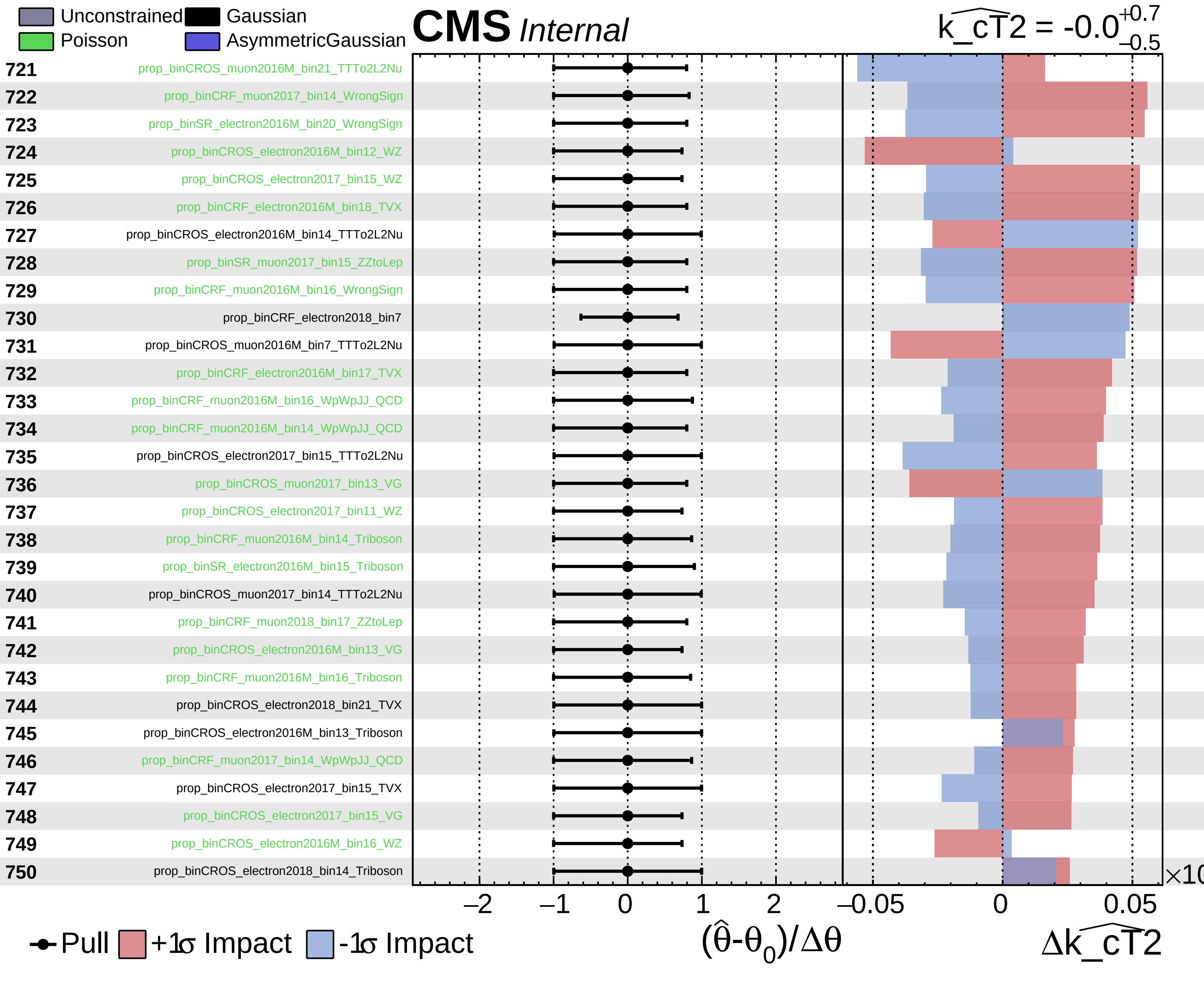


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+0.7}_{-0.5}$
631 prop_binCRF_muon2018_bin17_TVX
632 prop_binCROS_muon2017_bin4
633 prop_binCRF_electron2017_bin17_WrongSign
634 prop_binCRF_muon2017_bin11_TVX
635 prop_binCRF_electron2016M_bin20
636 prop_binSR_electron2017_bin15_WrongSign
637 prop_binCRF_muon2017_bin7
638 prop_binCROS_muon2017_bin13_TTTo2L2Nu
639 prop_binSR_muon2017_bin4
640 prop_binCRF_muon2018_bin8
641 prop_binCROS_muon2017_bin11_TTTo2L2Nu
642 prop_binCROS_muon2016M_bin9_TTTo2L2Nu
643 prop_binCRF_muon2017_bin11_WrongSign
644 prop_binCRF_electron2017_bin10
645 prop_binSR_electron2017_bin3
646 prop_binCROS_muon2017_bin17_Fake
647 prop_binCROS_electron2016M_bin13_Fake
648 prop_binCROS_electron2016M_bin14_Fake
649 prop_binCRF_electron2018_bin12_sm
650 prop_binCRF_electron2017_bin9
651 prop_binCRF_electron2016M_bin16_WrongSign
652 prop_binSR_electron2016M_bin21_ZZtoLep
653 prop_binCROS_electron2017_bin15_Fake
654 prop_binCRF_muon2016M_bin14_WrongSign
655 prop_binCRF_electron2016M_bin7
656 prop_binSR_muon2018_bin0
657 prop_binCRF_muon2016M_bin14_TVX
658 prop_binCRTT_muon2017_bin19
659 prop_binCRF_electron2018_bin9
660 prop_binCROS_muon2016M_bin21_WrongSign
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

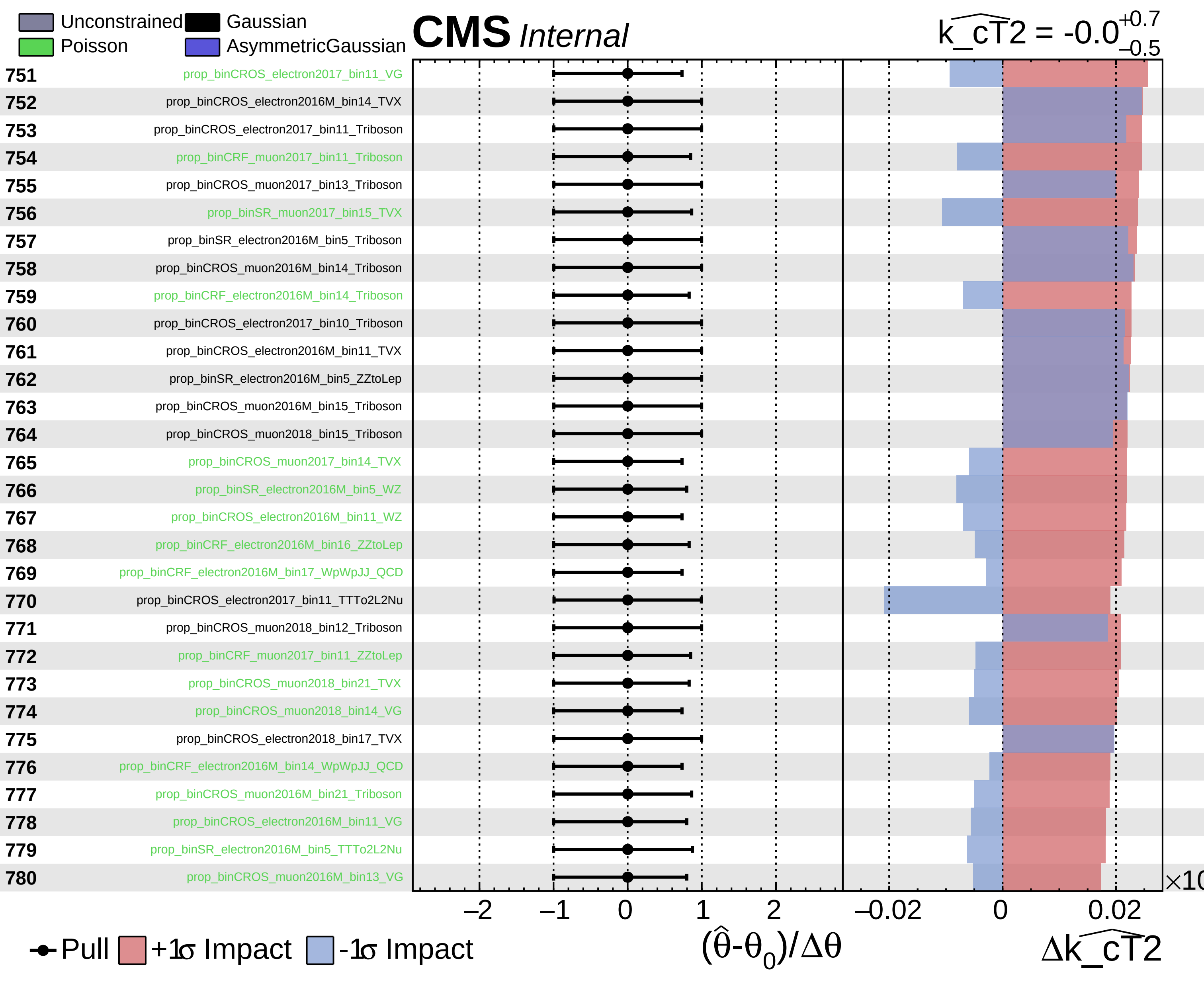

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
661 prop_binCRTT_electron2017_bin21
662 prop_binCROS_electron2017_bin10_TTTo2L2Nu
663 prop_binCROS_muon2018_bin20
664 prop_binSR_electron2016M_bin21_WrongSign
665 prop_binSR_muon2017_bin0
666 prop_binCROS_electron2017_bin21_TTTo2L2Nu
667 prop_binCROS_electron2016M_bin16_Fake
668 prop_binCROS_muon2017_bin21_TTTo2L2Nu
669 prop_binSR_electron2017_bin20_WrongSign
670 prop_binCROS_muon2017_bin13_Fake
671 prop_binCRF_muon2016M_bin16_TVX
672 prop_binCRTT_muon2016M_bin21
673 lep_2016M
674 prop_binSR_electron2016M_bin20_ZZtoLep
675 prop_binCROS_electron2016M_bin16_TTTo2L2Nu
676 prop_binCROS_electron2018_bin13_TTTo2L2Nu
677 prop_binCROS_muon2017_bin17_TTTo2L2Nu
678 prop_binCROS_muon2017_bin11_VG
679 prop_binSR_electron2016M_bin15_TVX
680 prop_binSR_electron2017_bin15_Triboson
681 prop_binCROS_electron2017_bin13_TTTo2L2Nu
682 prop_binCROS_electron2016M_bin20_TTTo2L2Nu
683 prop_binSR_muon2016M_bin4
684 prop_binCRF_muon2017_bin14_TTTo2L2Nu
685 prop_binCROS_electron2017_bin16_TTTo2L2Nu
686 prop_binCRF_electron2016M_bin16_Triboson
687 prop_binCRTT_electron2018_bin21
688 prop_binCROS_muon2018_bin12_VG
689 prop_binCRF_electron2016M_bin16_TVX
690 prop_binCRF_electron2016M_bin14_WrongSign
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



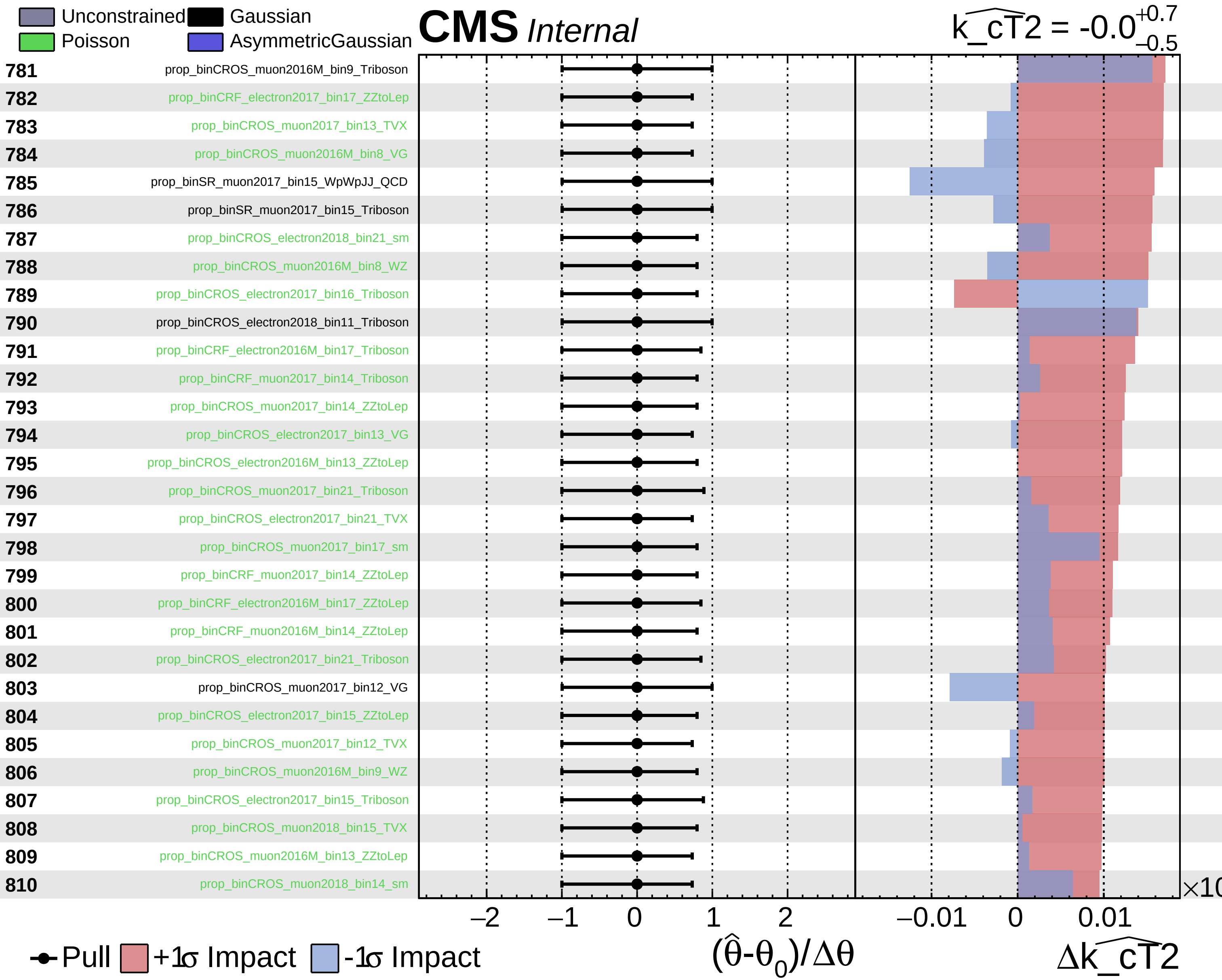
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+0.7}_{-0.5}$
691 prop_binSR_muon2017_bin15_TTTo2L2Nu
692 prop_binCROS_muon2018_bin12_WZ
693 prop_binSR_muon2017_bin19_WrongSign
694 prop_binCRF_electron2016M_bin14_TVX
695 prop_binCROS_electron2018_bin21_Triboson
696 prop_binCRF_electron2017_bin18_WpWpJJ_QCD
697 prop_binCROS_muon2017_bin16_TTTo2L2Nu
698 prop_binSR_electron2016M_bin15_WpWpJJ_QCD
699 prop_binCRF_muon2018_bin17_WrongSign
700 prop_binSR_electron2017_bin2
701 prop_binSR_electron2016M_bin5_sm
702 prop_binSR_electron2017_bin20_ZZtoLep
703 prop_binCRF_muon2018_bin17_WpWpJJ_QCD
704 prop_binCROS_electron2018_bin13_VG
705 prop_binCROS_electron2016M_bin13_TTTo2L2Nu
706 prop_binCRF_electron2016M_bin16_WpWpJJ_QCD
707 prop_binCROS_muon2018_bin21_Triboson
708 prop_binCROS_electron2017_bin20
709 prop_binCRF_electron2017_bin17_WpWpJJ_QCD
710 prop_binCRF_electron2016M_bin15
711 prop_binCROS_muon2017_bin19
712 prop_binCRF_electron2016M_bin18_WpWpJJ_QCD
713 prop_binCROS_electron2016M_bin21_Triboson
714 prop_binCRF_muon2017_bin11_WpWpJJ_QCD
715 prop_binCROS_muon2018_bin16_VG
716 prop_binCRF_muon2018_bin17_Triboson
717 prop_binCRF_muon2016M_bin9
718 prop_binSR_electron2016M_bin15_WrongSign
719 prop_binCRF_muon2016M_bin10
720 prop_binCRF_muon2017_bin14_TVX
−2
−1
0
1
2
−0.1
−0.05
0
0.05
0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



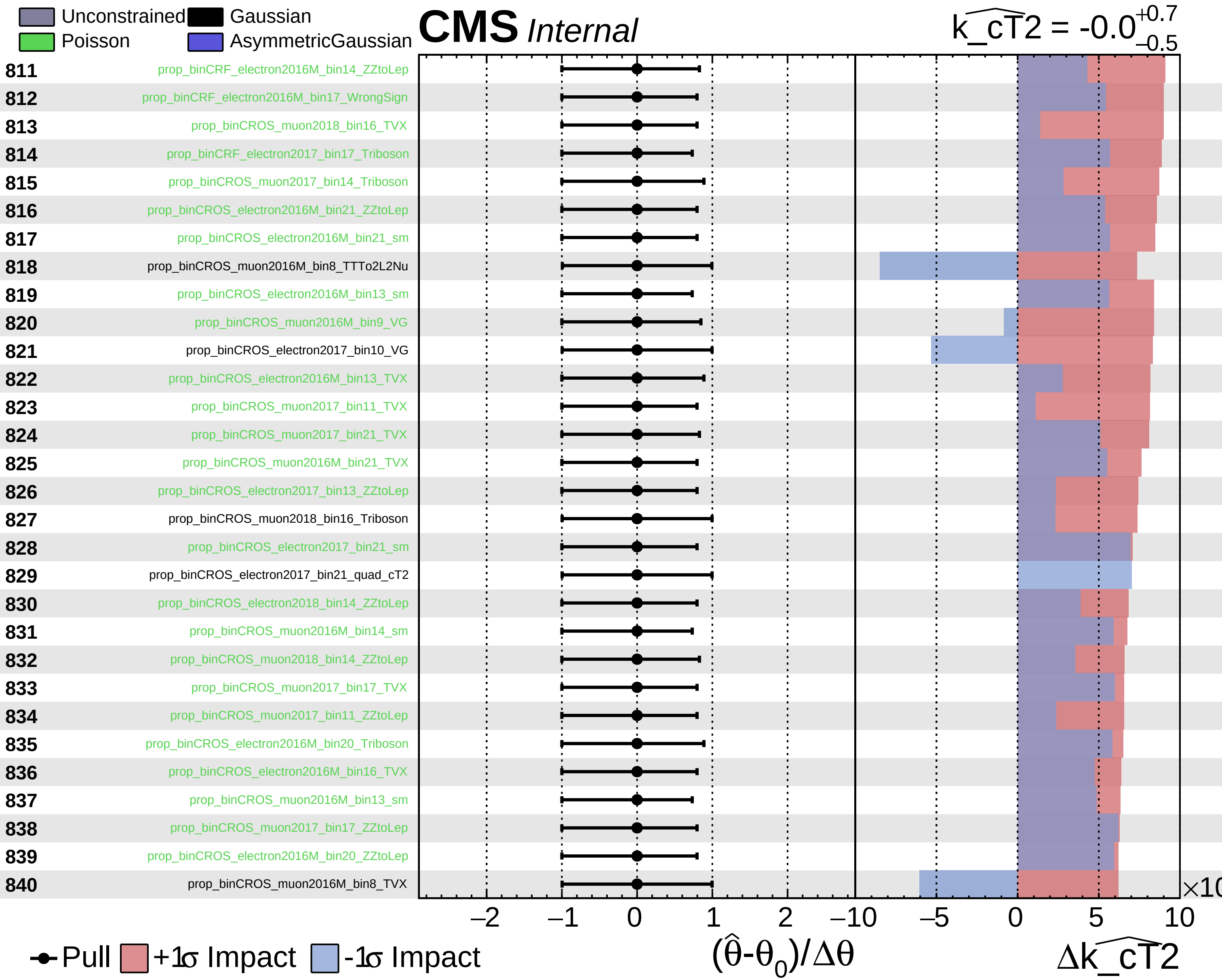
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0+0.7 −0.5
721 prop_binCROS_muon2016M_bin21_TTTo2L2Nu
722 prop_binCRF_muon2017_bin14_WrongSign
723 prop_binSR_electron2016M_bin20_WrongSign
724 prop_binCROS_electron2016M_bin12_WZ
725 prop_binCROS_electron2017_bin15_WZ
726 prop_binCRF_electron2016M_bin18_TVX
727 prop_binCROS_electron2016M_bin14_TTTo2L2Nu
728 prop_binSR_muon2017_bin15_ZZtoLep
729 prop_binCRF_muon2016M_bin16_WrongSign
730 prop_binCRF_electron2018_bin7
731 prop_binCROS_muon2016M_bin7_TTTo2L2Nu
732 prop_binCRF_electron2016M_bin17_TVX
733 prop_binCRF_muon2016M_bin16_WpWpJJ_QCD
734 prop_binCRF_muon2016M_bin14_WpWpJJ_QCD
735 prop_binCROS_electron2017_bin15_TTTo2L2Nu
736 prop_binCROS_muon2017_bin13_VG
737 prop_binCROS_electron2017_bin11_WZ
738 prop_binCRF_muon2016M_bin14_Triboson
739 prop_binSR_electron2016M_bin15_Triboson
740 prop_binCROS_muon2017_bin14_TTTo2L2Nu
741 prop_binCRF_muon2018_bin17_ZZtoLep
742 prop_binCROS_electron2016M_bin13_VG
743 prop_binCRF_muon2016M_bin16_Triboson
744 prop_binCROS_electron2018_bin21_TVX
745 prop_binCROS_electron2016M_bin13_Triboson
746 prop_binCRF_muon2017_bin14_WpWpJJ_QCD
747 prop_binCROS_electron2017_bin15_TVX
748 prop_binCROS_electron2017_bin15_VG
749 prop_binCROS_electron2016M_bin16_WZ
750 prop_binCROS_electron2018_bin14_Triboson
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT2

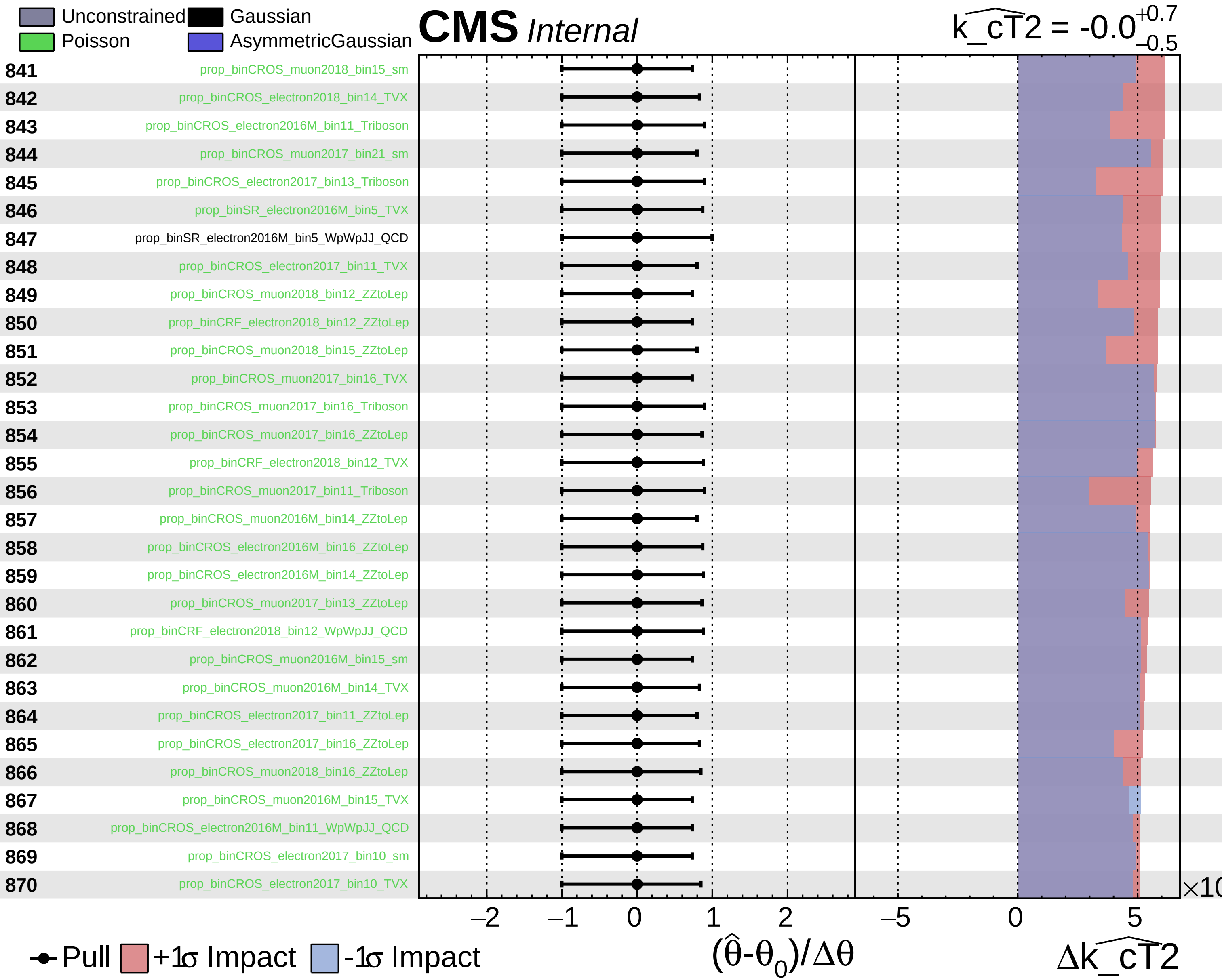


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
751 prop_binCROS_electron2017_bin11_VG
752 prop_binCROS_electron2016M_bin14_TVX
753 prop_binCROS_electron2017_bin11_Triboson
754 prop_binCRF_muon2017_bin11_Triboson
755 prop_binCROS_muon2017_bin13_Triboson
756 prop_binSR_muon2017_bin15_TVX
757 prop_binSR_electron2016M_bin5_Triboson
758 prop_binCROS_muon2016M_bin14_Triboson
759 prop_binCRF_electron2016M_bin14_Triboson
760 prop_binCROS_electron2017_bin10_Triboson
761 prop_binCROS_electron2016M_bin11_TVX
762 prop_binSR_electron2016M_bin5_ZZtoLep
763 prop_binCROS_muon2016M_bin15_Triboson
764 prop_binCROS_muon2018_bin15_Triboson
765 prop_binCROS_muon2017_bin14_TVX
766 prop_binSR_electron2016M_bin5_WZ
767 prop_binCROS_electron2016M_bin11_WZ
768 prop_binCRF_electron2016M_bin16_ZZtoLep
769 prop_binCRF_electron2016M_bin17_WpWpJJ_QCD
770 prop_binCROS_electron2017_bin11_TTTo2L2Nu
771 prop_binCROS_muon2018_bin12_Triboson
772 prop_binCRF_muon2017_bin11_ZZtoLep
773 prop_binCROS_muon2018_bin21_TVX
774 prop_binCROS_muon2018_bin14_VG
775 prop_binCROS_electron2018_bin17_TVX
776 prop_binCRF_electron2016M_bin14_WpWpJJ_QCD
777 prop_binCROS_muon2016M_bin21_Triboson
778 prop_binCROS_electron2016M_bin11_VG
779 prop_binSR_electron2016M_bin5_TTTo2L2Nu
780 prop_binCROS_muon2016M_bin13_VG
−2 −1 0 1 2
−0.02 0 0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

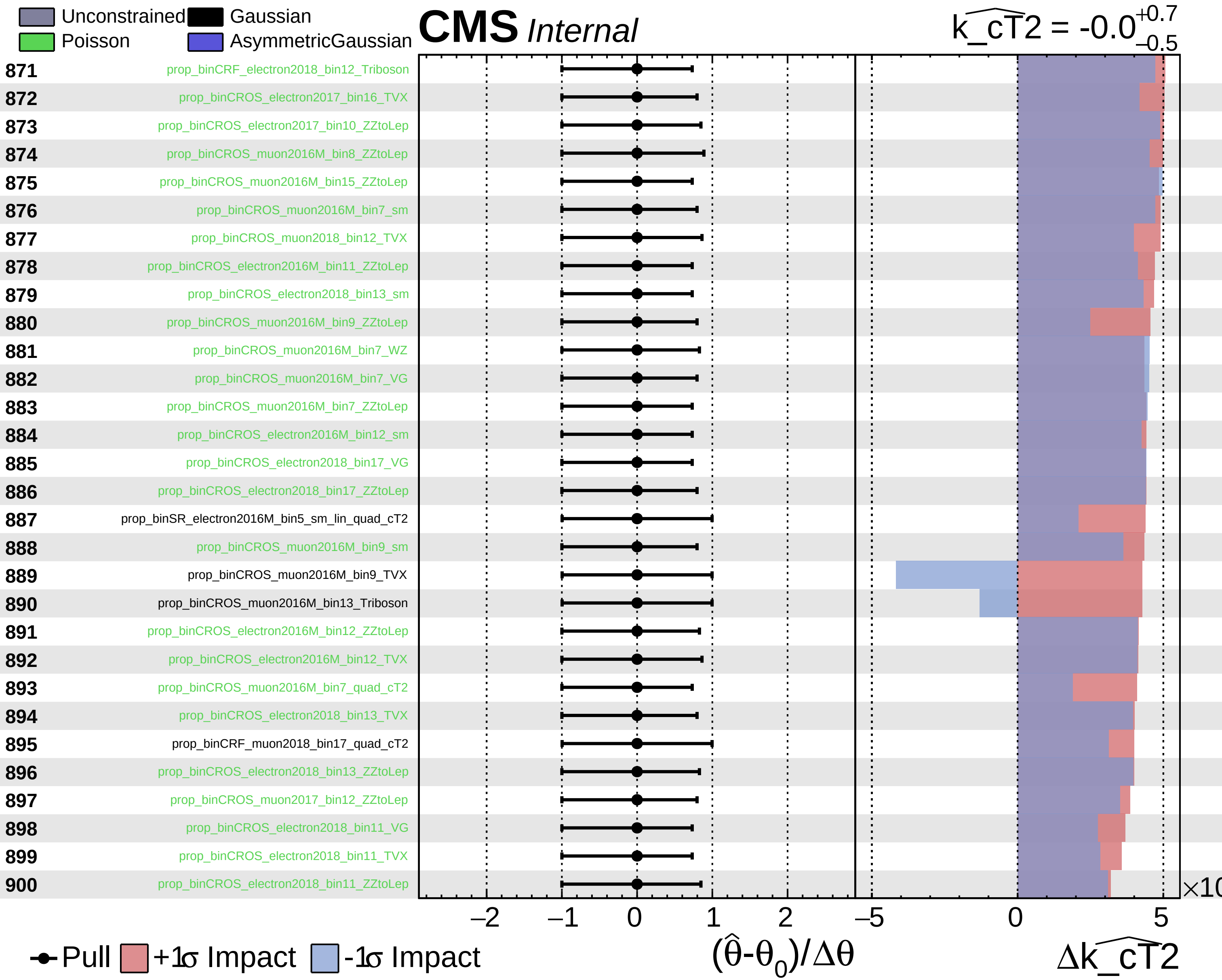


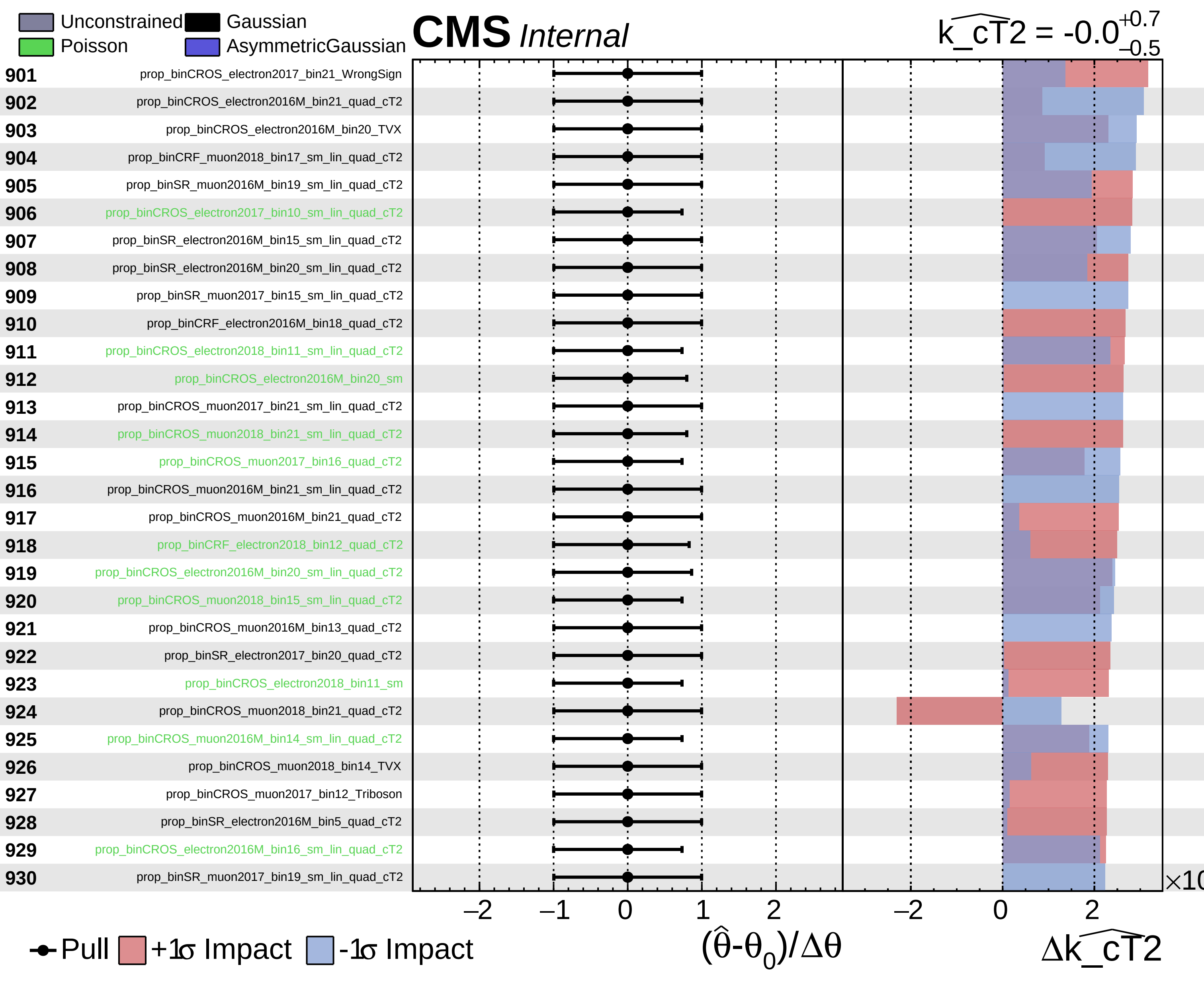
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+0.7}_{-0.5}$
781 prop_binCROS_muon2016M_bin9_Triboson
782 prop_binCRF_electron2017_bin17_ZZtoLep
783 prop_binCROS_muon2017_bin13_TVX
784 prop_binCROS_muon2016M_bin8_VG
785 prop_binSR_muon2017_bin15_WpWpJJ_QCD
786 prop_binSR_muon2017_bin15_Triboson
787 prop_binCROS_electron2018_bin21_sm
788 prop_binCROS_muon2016M_bin8_WZ
789 prop_binCROS_electron2017_bin16_Triboson
790 prop_binCROS_electron2018_bin11_Triboson
791 prop_binCRF_electron2016M_bin17_Triboson
792 prop_binCRF_muon2017_bin14_Triboson
793 prop_binCROS_muon2017_bin14_ZZtoLep
794 prop_binCROS_electron2017_bin13_VG
795 prop_binCROS_electron2016M_bin13_ZZtoLep
796 prop_binCROS_muon2017_bin21_Triboson
797 prop_binCROS_electron2017_bin21_TVX
798 prop_binCROS_muon2017_bin17_sm
799 prop_binCRF_muon2017_bin14_ZZtoLep
800 prop_binCRF_electron2016M_bin17_ZZtoLep
801 prop_binCRF_muon2016M_bin14_ZZtoLep
802 prop_binCROS_electron2017_bin21_Triboson
803 prop_binCROS_muon2017_bin12_VG
804 prop_binCROS_electron2017_bin15_ZZtoLep
805 prop_binCROS_muon2017_bin12_TVX
806 prop_binCROS_muon2016M_bin9_WZ
807 prop_binCROS_electron2017_bin15_Triboson
808 prop_binCROS_muon2018_bin15_TVX
809 prop_binCROS_muon2016M_bin13_ZZtoLep
810 prop_binCROS_muon2018_bin14_sm
−2
−1
0
1
2
−0.01
0
0.01
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
811 prop_binCRF_electron2016M_bin14_ZZtoLep
812 prop_binCRF_electron2016M_bin17_WrongSign
813 prop_binCROS_muon2018_bin16_TVX
814 prop_binCRF_electron2017_bin17_Triboson
815 prop_binCROS_muon2017_bin14_Triboson
816 prop_binCROS_electron2016M_bin21_ZZtoLep
817 prop_binCROS_electron2016M_bin21_sm
818 prop_binCROS_muon2016M_bin8_TTTo2L2Nu
819 prop_binCROS_electron2016M_bin13_sm
820 prop_binCROS_muon2016M_bin9_VG
821 prop_binCROS_electron2017_bin10_VG
822 prop_binCROS_electron2016M_bin13_TVX
823 prop_binCROS_muon2017_bin11_TVX
824 prop_binCROS_muon2017_bin21_TVX
825 prop_binCROS_muon2016M_bin21_TVX
826 prop_binCROS_electron2017_bin13_ZZtoLep
827 prop_binCROS_muon2018_bin16_Triboson
828 prop_binCROS_electron2017_bin21_sm
829 prop_binCROS_electron2017_bin21_quad_cT2
830 prop_binCROS_electron2018_bin14_ZZtoLep
831 prop_binCROS_muon2016M_bin14_sm
832 prop_binCROS_muon2018_bin14_ZZtoLep
833 prop_binCROS_muon2017_bin17_TVX
834 prop_binCROS_muon2017_bin11_ZZtoLep
835 prop_binCROS_electron2016M_bin20_Triboson
836 prop_binCROS_electron2016M_bin16_TVX
837 prop_binCROS_muon2016M_bin13_sm
838 prop_binCROS_muon2017_bin17_ZZtoLep
839 prop_binCROS_electron2016M_bin20_ZZtoLep
840 prop_binCROS_muon2016M_bin8_TVX
−2 −1 0 1 2
−10 −5 0 5 10
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

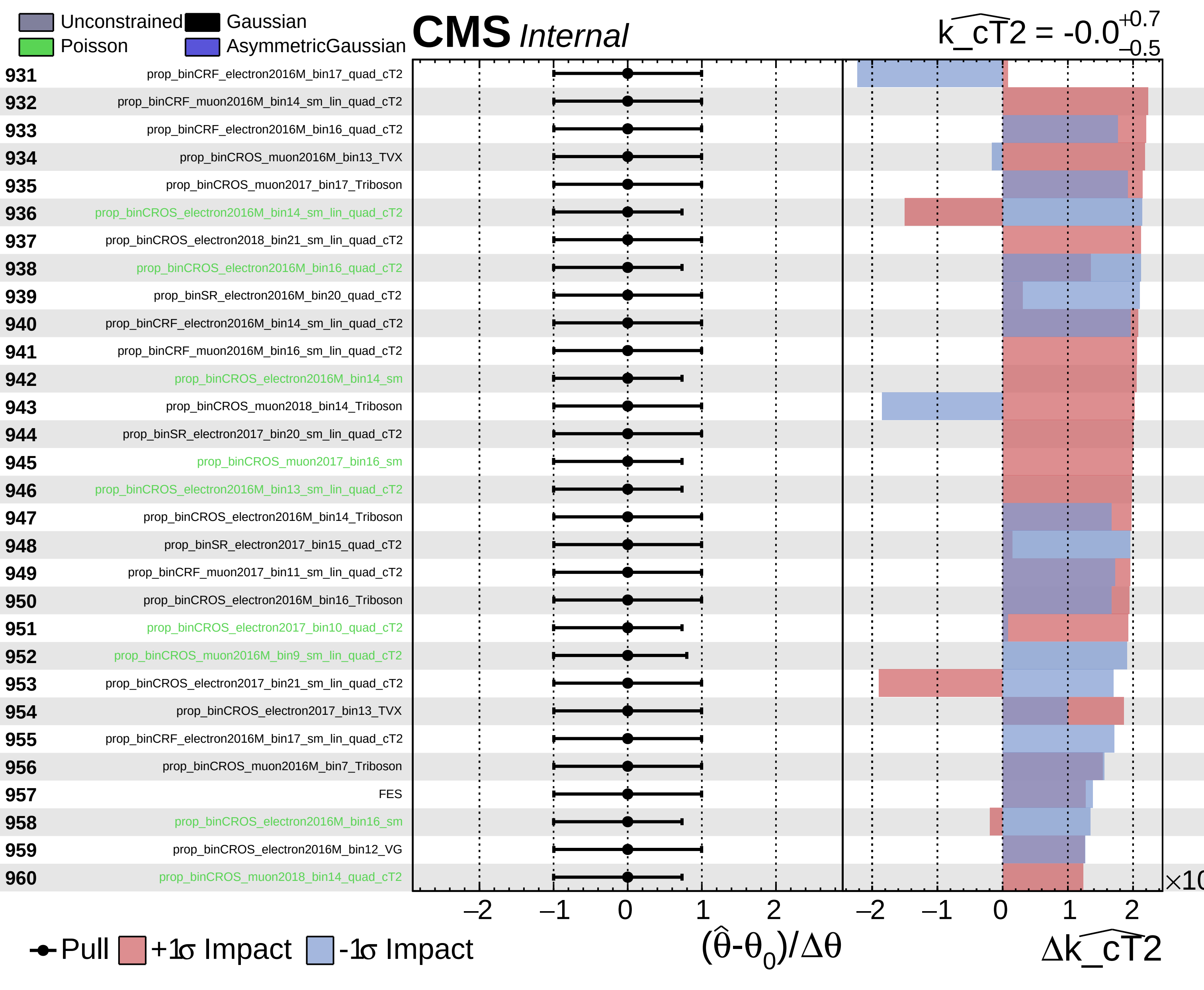


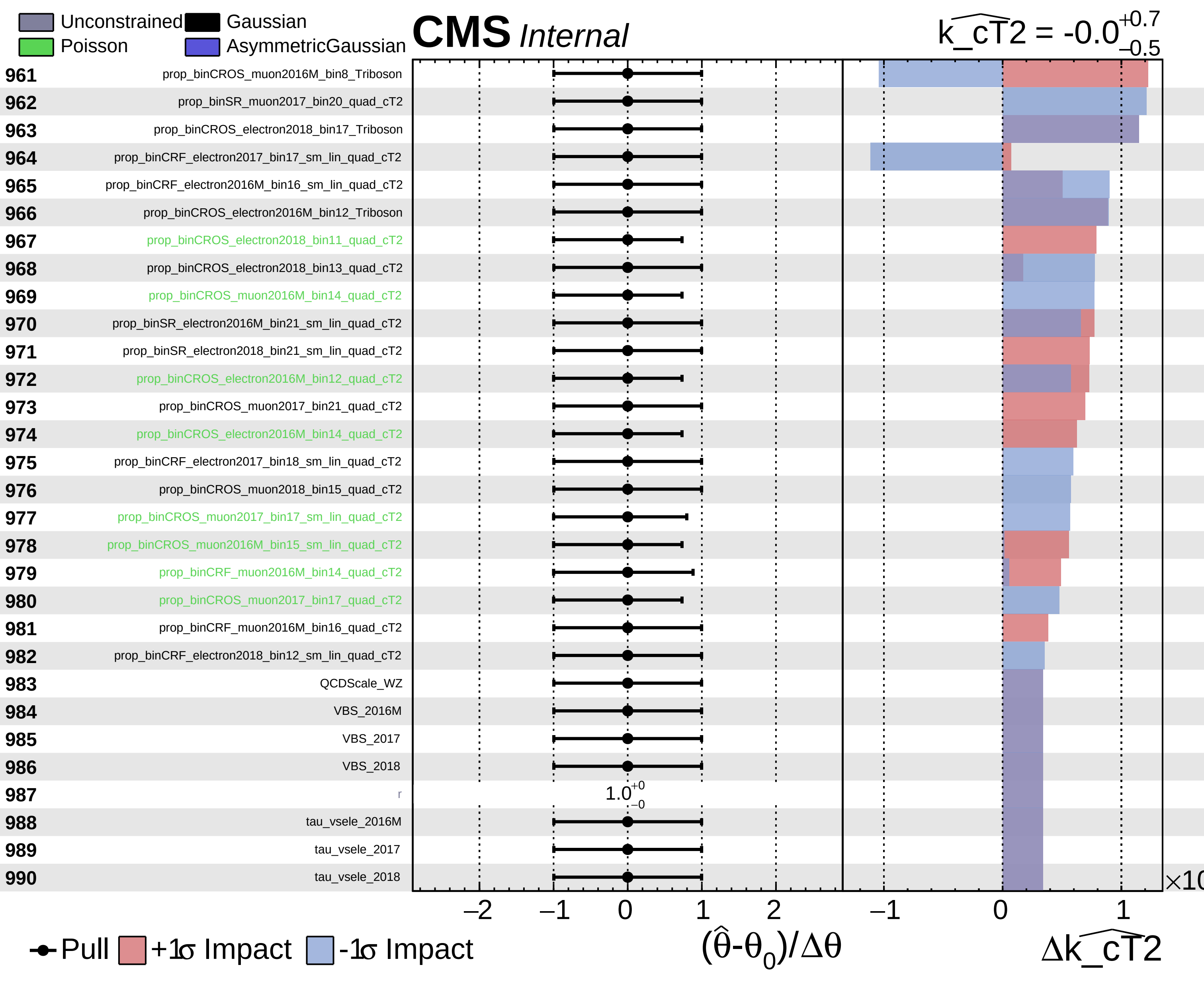
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
841 prop_binCROS_muon2018_bin15_sm
842 prop_binCROS_electron2018_bin14_TVX
843 prop_binCROS_electron2016M_bin11_Triboson
844 prop_binCROS_muon2017_bin21_sm
845 prop_binCROS_electron2017_bin13_Triboson
846 prop_binSR_electron2016M_bin5_TVX
847 prop_binSR_electron2016M_bin5_WpWpJJ_QCD
848 prop_binCROS_electron2017_bin11_TVX
849 prop_binCROS_muon2018_bin12_ZZtoLep
850 prop_binCRF_electron2018_bin12_ZZtoLep
851 prop_binCROS_muon2018_bin15_ZZtoLep
852 prop_binCROS_muon2017_bin16_TVX
853 prop_binCROS_muon2017_bin16_Triboson
854 prop_binCROS_muon2017_bin16_ZZtoLep
855 prop_binCRF_electron2018_bin12_TVX
856 prop_binCROS_muon2017_bin11_Triboson
857 prop_binCROS_muon2016M_bin14_ZZtoLep
858 prop_binCROS_electron2016M_bin16_ZZtoLep
859 prop_binCROS_electron2016M_bin14_ZZtoLep
860 prop_binCROS_muon2017_bin13_ZZtoLep
861 prop_binCRF_electron2018_bin12_WpWpJJ_QCD
862 prop_binCROS_muon2016M_bin15_sm
863 prop_binCROS_muon2016M_bin14_TVX
864 prop_binCROS_electron2017_bin11_ZZtoLep
865 prop_binCROS_electron2017_bin16_ZZtoLep
866 prop_binCROS_muon2018_bin16_ZZtoLep
867 prop_binCROS_muon2016M_bin15_TVX
868 prop_binCROS_electron2016M_bin11_WpWpJJ_QCD
869 prop_binCROS_electron2017_bin10_sm
870 prop_binCROS_electron2017_bin10_TVX
−2 −1 0 1 2
−5 0 5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
871 prop_binCRF_electron2018_bin12_Triboson
872 prop_binCROS_electron2017_bin16_TVX
873 prop_binCROS_electron2017_bin10_ZZtoLep
874 prop_binCROS_muon2016M_bin8_ZZtoLep
875 prop_binCROS_muon2016M_bin15_ZZtoLep
876 prop_binCROS_muon2016M_bin7_sm
877 prop_binCROS_muon2018_bin12_TVX
878 prop_binCROS_electron2016M_bin11_ZZtoLep
879 prop_binCROS_electron2018_bin13_sm
880 prop_binCROS_muon2016M_bin9_ZZtoLep
881 prop_binCROS_muon2016M_bin7_WZ
882 prop_binCROS_muon2016M_bin7_VG
883 prop_binCROS_muon2016M_bin7_ZZtoLep
884 prop_binCROS_electron2016M_bin12_sm
885 prop_binCROS_electron2018_bin17_VG
886 prop_binCROS_electron2018_bin17_ZZtoLep
887 prop_binSR_electron2016M_bin5_sm_lin_quad_cT2
888 prop_binCROS_muon2016M_bin9_sm
889 prop_binCROS_muon2016M_bin9_TVX
890 prop_binCROS_muon2016M_bin13_Triboson
891 prop_binCROS_electron2016M_bin12_ZZtoLep
892 prop_binCROS_electron2016M_bin12_TVX
893 prop_binCROS_muon2016M_bin7_quad_cT2
894 prop_binCROS_electron2018_bin13_TVX
895 prop_binCRF_muon2018_bin17_quad_cT2
896 prop_binCROS_electron2018_bin13_ZZtoLep
897 prop_binCROS_muon2017_bin12_ZZtoLep
898 prop_binCROS_electron2018_bin11_VG
899 prop_binCROS_electron2018_bin11_TVX
900 prop_binCROS_electron2018_bin11_ZZtoLep
−2
−1
0
1
2
−5
0
5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
901 prop_binCROS_electron2017_bin21_WrongSign
902 prop_binCROS_electron2016M_bin21_quad_cT2
903 prop_binCROS_electron2016M_bin20_TVX
904 prop_binCRF_muon2018_bin17_sm_lin_quad_cT2
905 prop_binSR_muon2016M_bin19_sm_lin_quad_cT2
906 prop_binCROS_electron2017_bin10_sm_lin_quad_cT2
907 prop_binSR_electron2016M_bin15_sm_lin_quad_cT2
908 prop_binSR_electron2016M_bin20_sm_lin_quad_cT2
909 prop_binSR_muon2017_bin15_sm_lin_quad_cT2
910 prop_binCRF_electron2016M_bin18_quad_cT2
911 prop_binCROS_electron2018_bin11_sm_lin_quad_cT2
912 prop_binCROS_electron2016M_bin20_sm
913 prop_binCROS_muon2017_bin21_sm_lin_quad_cT2
914 prop_binCROS_muon2018_bin21_sm_lin_quad_cT2
915 prop_binCROS_muon2017_bin16_quad_cT2
916 prop_binCROS_muon2016M_bin21_sm_lin_quad_cT2
917 prop_binCROS_muon2016M_bin21_quad_cT2
918 prop_binCRF_electron2018_bin12_quad_cT2
919 prop_binCROS_electron2016M_bin20_sm_lin_quad_cT2
920 prop_binCROS_muon2018_bin15_sm_lin_quad_cT2
921 prop_binCROS_muon2016M_bin13_quad_cT2
922 prop_binSR_electron2017_bin20_quad_cT2
923 prop_binCROS_electron2018_bin11_sm
924 prop_binCROS_muon2018_bin21_quad_cT2
925 prop_binCROS_muon2016M_bin14_sm_lin_quad_cT2
926 prop_binCROS_muon2018_bin14_TVX
927 prop_binCROS_muon2017_bin12_Triboson
928 prop_binSR_electron2016M_bin5_quad_cT2
929 prop_binCROS_electron2016M_bin16_sm_lin_quad_cT2
930 prop_binSR_muon2017_bin19_sm_lin_quad_cT2
−2
−1
0
1
2
−2
0
2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+0.7}_{-0.5}$
931 prop_binCRF_electron2016M_bin17_quad_cT2
932 prop_binCRF_muon2016M_bin14_sm_lin_quad_cT2
933 prop_binCRF_electron2016M_bin16_quad_cT2
934 prop_binCROS_muon2016M_bin13_TVX
935 prop_binCROS_muon2017_bin17_Triboson
936 prop_binCROS_electron2016M_bin14_sm_lin_quad_cT2
937 prop_binCROS_electron2018_bin21_sm_lin_quad_cT2
938 prop_binCROS_electron2016M_bin16_quad_cT2
939 prop_binSR_electron2016M_bin20_quad_cT2
940 prop_binCRF_electron2016M_bin14_sm_lin_quad_cT2
941 prop_binCRF_muon2016M_bin16_sm_lin_quad_cT2
942 prop_binCROS_electron2016M_bin14_sm
943 prop_binCROS_muon2018_bin14_Triboson
944 prop_binSR_electron2017_bin20_sm_lin_quad_cT2
945 prop_binCROS_muon2017_bin16_sm
946 prop_binCROS_electron2016M_bin13_sm_lin_quad_cT2
947 prop_binCROS_electron2016M_bin14_Triboson
948 prop_binSR_electron2017_bin15_quad_cT2
949 prop_binCRF_muon2017_bin11_sm_lin_quad_cT2
950 prop_binCROS_electron2016M_bin16_Triboson
951 prop_binCROS_electron2017_bin10_quad_cT2
952 prop_binCROS_muon2016M_bin9_sm_lin_quad_cT2
953 prop_binCROS_electron2017_bin21_sm_lin_quad_cT2
954 prop_binCROS_electron2017_bin13_TVX
955 prop_binCRF_electron2016M_bin17_sm_lin_quad_cT2
956 prop_binCROS_muon2016M_bin7_Triboson
957 FES
958 prop_binCROS_electron2016M_bin16_sm
959 prop_binCROS_electron2016M_bin12_VG
960 prop_binCROS_muon2018_bin14_quad_cT2
−2 −1 0 1 2
$(\hat{\theta}-\theta_0)/\Delta\theta$
−2 −1 0 1 2
×10
$\Delta\widehat{k\_cT2}$
Pull
+1σ Impact
-1σ Impact


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
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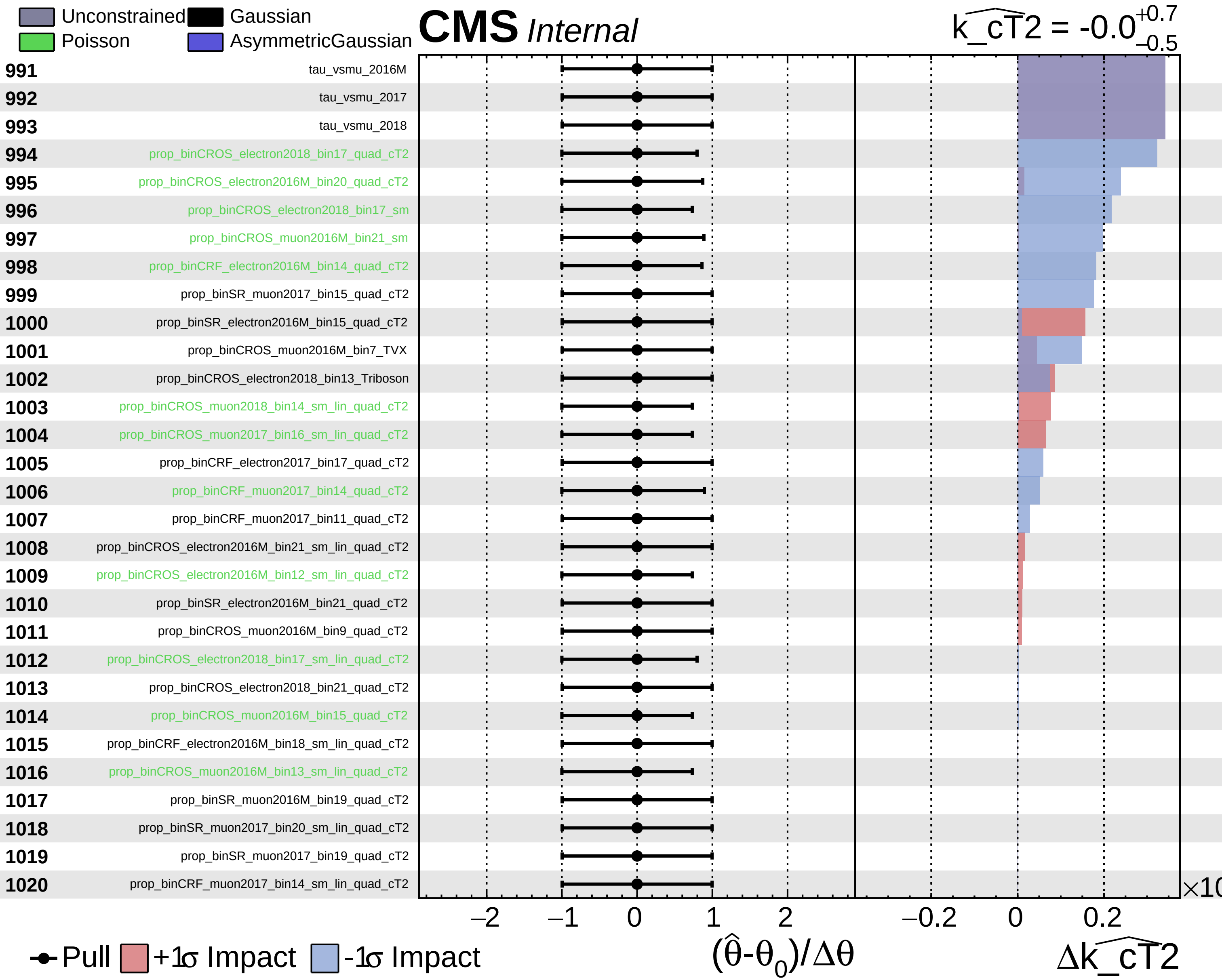
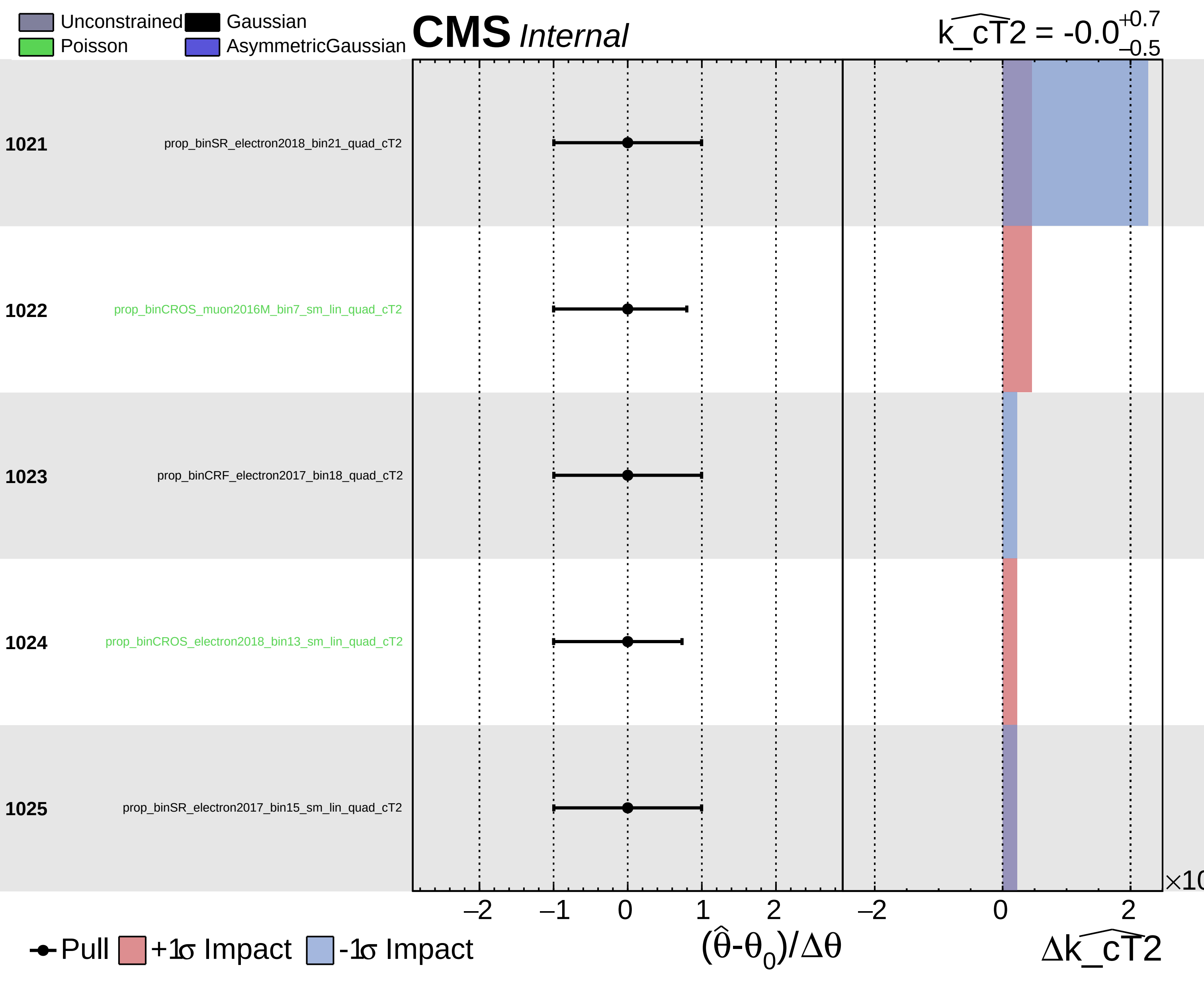
962 prop_binSR_muon2017_bin20_quad_cT2
963 prop_binCROS_electron2018_bin17_Triboson
964 prop_binCRF_electron2017_bin17_sm_lin_quad_cT2
965 prop_binCRF_electron2016M_bin16_sm_lin_quad_cT2
966 prop_binCROS_electron2016M_bin12_Triboson
967 prop_binCROS_electron2018_bin11_quad_cT2
968 prop_binCROS_electron2018_bin13_quad_cT2
969 prop_binCROS_muon2016M_bin14_quad_cT2
970 prop_binSR_electron2016M_bin21_sm_lin_quad_cT2
971 prop_binSR_electron2018_bin21_sm_lin_quad_cT2
972 prop_binCROS_electron2016M_bin12_quad_cT2
973 prop_binCROS_muon2017_bin21_quad_cT2
974 prop_binCROS_electron2016M_bin14_quad_cT2
975 prop_binCRF_electron2017_bin18_sm_lin_quad_cT2
976 prop_binCROS_muon2018_bin15_quad_cT2
977 prop_binCROS_muon2017_bin17_sm_lin_quad_cT2
978 prop_binCROS_muon2016M_bin15_sm_lin_quad_cT2
979 prop_binCRF_muon2016M_bin14_quad_cT2
980 prop_binCROS_muon2017_bin17_quad_cT2
981 prop_binCRF_muon2016M_bin16_quad_cT2
982 prop_binCRF_electron2018_bin12_sm_lin_quad_cT2
983 QCDScale_WZ
984 VBS_2016M
985 VBS_2017
986 VBS_2018
987 r
1.0 +0 −0
988 tau_vsele_2016M
989 tau_vsele_2017
990 tau_vsele_2018
−2 −1 0 1 2
−1 0 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
991 tau_vsmu_2016M
992 tau_vsmu_2017
993 tau_vsmu_2018
994 prop_binCROS_electron2018_bin17_quad_cT2
995 prop_binCROS_electron2016M_bin20_quad_cT2
996 prop_binCROS_electron2018_bin17_sm
997 prop_binCROS_muon2016M_bin21_sm
998 prop_binCRF_electron2016M_bin14_quad_cT2
999 prop_binSR_muon2017_bin15_quad_cT2
1000 prop_binSR_electron2016M_bin15_quad_cT2
1001 prop_binCROS_muon2016M_bin7_TVX
1002 prop_binCROS_electron2018_bin13_Triboson
1003 prop_binCROS_muon2018_bin14_sm_lin_quad_cT2
1004 prop_binCROS_muon2017_bin16_sm_lin_quad_cT2
1005 prop_binCRF_electron2017_bin17_quad_cT2
1006 prop_binCRF_muon2017_bin14_quad_cT2
1007 prop_binCRF_muon2017_bin11_quad_cT2
1008 prop_binCROS_electron2016M_bin21_sm_lin_quad_cT2
1009 prop_binCROS_electron2016M_bin12_sm_lin_quad_cT2
1010 prop_binSR_electron2016M_bin21_quad_cT2
1011 prop_binCROS_muon2016M_bin9_quad_cT2
1012 prop_binCROS_electron2018_bin17_sm_lin_quad_cT2
1013 prop_binCROS_electron2018_bin21_quad_cT2
1014 prop_binCROS_muon2016M_bin15_quad_cT2
1015 prop_binCRF_electron2016M_bin18_sm_lin_quad_cT2
1016 prop_binCROS_muon2016M_bin13_sm_lin_quad_cT2
1017 prop_binSR_muon2016M_bin19_quad_cT2
1018 prop_binSR_muon2017_bin20_sm_lin_quad_cT2
1019 prop_binSR_muon2017_bin19_quad_cT2
1020 prop_binCRF_muon2017_bin14_sm_lin_quad_cT2
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +0.7 −0.5
1021
prop_binSR_electron2018_bin21_quad_cT2
1022
prop_binCROS_muon2016M_bin7_sm_lin_quad_cT2
1023
prop_binCRF_electron2017_bin18_quad_cT2
1024
prop_binCROS_electron2018_bin13_sm_lin_quad_cT2
1025
prop_binSR_electron2017_bin15_sm_lin_quad_cT2
−2 −1 0 1 2
−2 0 2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2